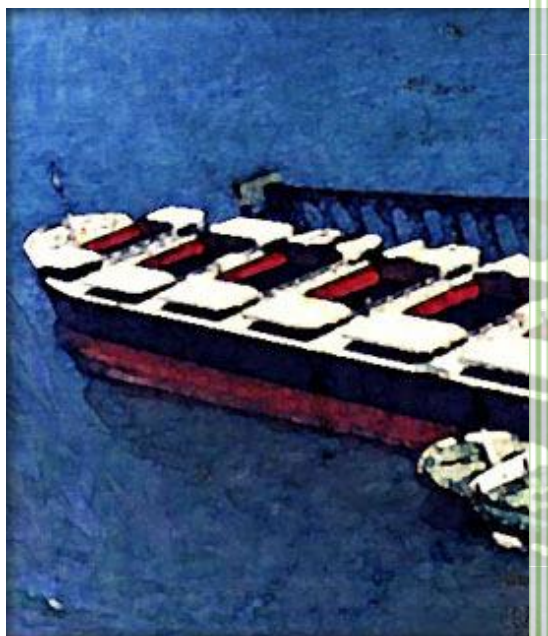


BUNKER QUANTITY SURVEY



BSOL SYSTEMS PVT LTD.

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Bunker Quantity Survey

1 Introduction:

1.1 About Bunker Quantity Survey

The Bunker Quantity Survey includes the processes and procedures, requirements, roles, and responsibilities of all parties concerned, for the delivery of bunkers by bunker tankers in the Port, including documentation, equipment standards and verification processes during a bunkering operation. It covers pre-delivery, actual delivery and post-delivery checks and documentation.

1.2 Why bunker survey?

A bunker survey can be a valuable loss control tool in locating cargo losses where the ship to shore difference is greater than normal. If there is an unaccountable gain in bunker tanks or bunker volumes that cannot be reconciled, a bunker survey can help identify the causes. Even though the vessel's fuel system is intrinsically separate from the cargo system, intentional or unintentional cargo diversion can take place and should be monitored especially where cargo can be used as bunker fuel. Comparisons are made to what the bunker volumes should be based on the last fuel intake and normal consumption verses what was found.

1.3 Steps for Conducting a Bunker Survey

When a bunker survey is requested, it is necessary to take and record all innages/ullages and pull representative samples from all bunker tanks. All measurements are taken prior to the cargo transfer and then again prior to the vessel's departure. The vessel's bunker volumes are calculated again on port arrival and at departure. The difference between these two volumes will be the vessel's fuel consumption at dock. Listed below are the procedural steps required to properly conduct a bunker survey:

- Identifies and records the number of Bunker Tanks, Depth of each Tank, Reference Height, and Measuring Method (Ullage or Sounding).
- Obtain and record the previous Bunker Report, time, and place the bunkers were supplied along with the oil density.
- Records the quantity of bunker onboard when the ship arrived (arrival Condition) as per Engine Logbook.
- Check the Ship's Draft Marks to get the Ship Trim for Trim correction and check the ship Inclination or Listing for List Correction.
- Gauge each Bunker Tank and record the level of Oil on the tank along with the oil temperature.
- Sample and water cut each bunker tank.
- Calculate the volumes in each tank using the Ship's Trim and List, Tank Quantity Table (provided by Ship's Chief Engineer), Oil Density and Temperature, ASTM Table for Volume Correction Factor (VCF).
- Review the data obtained and report and discuss any irregularities.

1.4 Conclusion

Bunker pricing and fuel quality are critical components to a vessel's operating costs. Fuel cost can be as much as 60% of a vessel's operating budget. Off specification fuel can potentially damage vessel engines and boilers. A bunker survey is relatively cheap insurance to deter product diversion. A detailed bunker survey will confirm fuel delivery and report bunker fuel shortages. If there is an unaccountable gain in bunker tanks or bunker volumes cannot be reconciled with bunker consumption, Note of Protests are issued, and an investigation can identify the causes.

2 Types of Users and Roles:

2.1 Admin:

- a. Admin cannot create survey.
- b. Admin can create/edit user.
- c. Admin can see all the Ongoing surveys and Published surveys.
- d. Admin has access to Master details and create/modify Master Data.

2.2 User:

- a. User is only able to read the Survey Data and Master data.

3 Bunker Quantity Survey Application High Level Workflow:

- Survey Company creates New Survey and assign it to Surveyor in Web Application (Central Server). (Status is set to Open)
- Admin conducts bunker survey and gathers all the information in the forms.
- Once the Survey is completed, it can be verified for the data and final report is sent to the Bunker Buyer (Customer).
- Once the Survey is submitted, the Survey will be closed and listed under Published Surveys.

3.1 Home Page:

After Logging into the application, User will be redirected to dashboard as shown in the following screen:

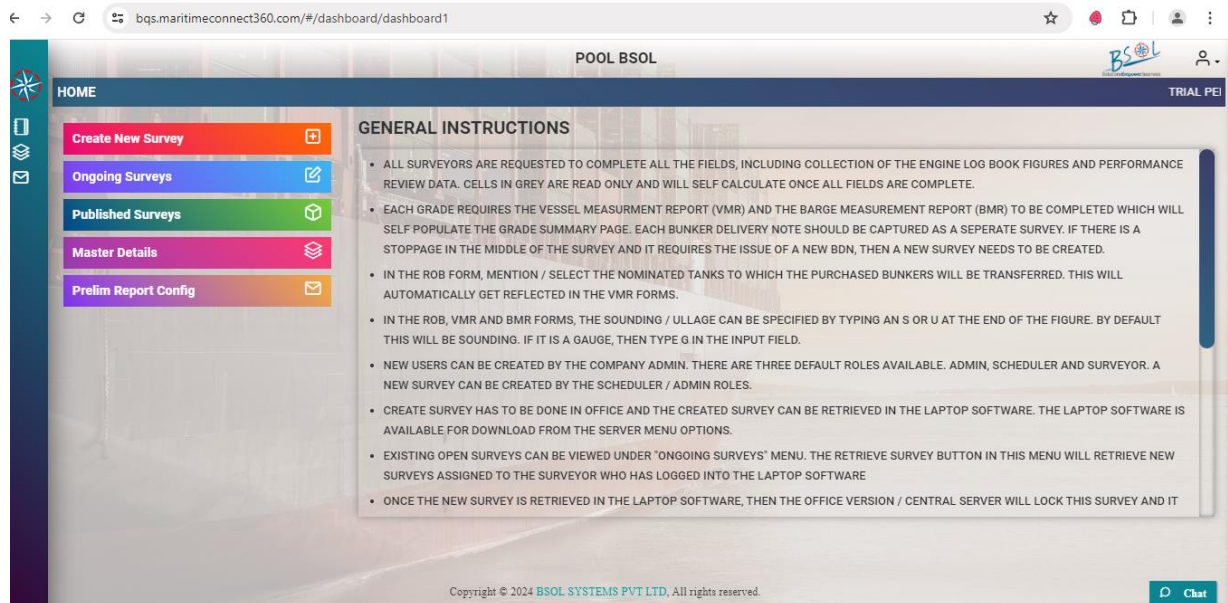


Figure 1 dashboard.

➤ We can see the following options in dashboard.

- General Instructions.
- Create New Survey.
- Ongoing Surveys.
- Published Surveys.
- Master Details.
- Prelim Report Config

Note:

- Depending on the type of user logged in to BQS application the features are displayed in Dashboard page.

3.2 Surveys:

- Examine and record the area and features of tanks.
- In this, we can see Create Survey, Ongoing Surveys, Published Surveys.
- All the created surveys will be listed under Ongoing Surveys as shown in the following screen:

POOL BSOL													
ONGOING SURVEYS													
Vessel	Client	Surveyor	Captain	Chief Engineer	Port	Location	Grade	Nominated Qty	Survey Date	Delivery Type	Survey ID	Status	Actions
Vessel	Client	Surveyor	Captain	Chief Engin	Port	Location	Grade	Nominated Q	Survey Da	Delivery Ty	Survey ID	Select	
TTTT	BUNGA CHANG	savita karagi	K L TRETET	CE MARGRATE	TESTS	CHINA	LSMGO	0.000	11-Jun-2024	Physical Gauge	13463	Open	
TESTT	BHAIRAVI NORDOCEAN GAZI TSDSKDFBDS FFDSMFNDSFNMDSF	krishnappa	AB JOSEPH	FU XIANG KAI	TESTT	MUMABI	VLSFO	342.000	05-Jun-2024	Physical Gauge	13505	Open	
TESTT	OLDENDORFF CARRIERS GMBH & CO. KG / NA	savita karagi	ROMEO E GATCHALIAN	JOKO PURWOTO	SINGAPORE	TEST	VLSFO	112.340	31-May-2024	Physical Gauge	13490	Open	
STAR	NORD SHIP	savita karagi	ROMEO E GATCHALIAN	JOKO PURWOTO	SINGAPORE	SINGAPORE	VLSFO	321.800	30-May-2024	Physical Gauge	13477	Open	
TEST	TESTS	savita karagi	K L TRETER	CE MARGRATE		SINGAPORE	VLSFO	554.000	30-May-2024	Physical Gauge	13489	Open	
TEEST	TEST	Lavanya	TEST CAPTAIN	7YHFGH		TEST	LSMGO	100.000	29-May-2024	MFM	13476	Open	
STAR	HAMBURG	savita karagi	AB JOSEPH	AGARWAL DAVID			VLSFO	0.000	24-May-2024	MFM	13462	Open	
TestVessel1	TRANSASIA BIO SSS	savita karagi	DON GURSILO	FU XIANG KAI			LSMGO	0.000	24-May-2024	Physical Gauge	13464	Open	

Figure 2 On going survey list

- Once the Survey is Submitted, it will be listed under Published Surveys as Shown in the following screen:

POOL BSOL										
PUBLISHED SURVEYS										
Vessel	Client	Surveyor	Captain	Chief Engineer	Port	Location	Grade	Nominated Qty	Survey Date	Survey ID
Vessel	Client	Surveyor	Captain	Chief Engineer	Port	Location	Grade	Nominated Q	Survey Date	Survey ID
STAR	TEST	Lavanya	TEST CAPTAIN	TEST	TESTS	SINGAPORE	LSMGO	455.000	05-Jun-2024	13508
TestVessel1		savita karagi					VLSFO	0.000	30-May-2024	13488
TEST	TEST	savita karagi	TEST	TEST	SINGAPORE	SINGAPORE	VLSFO	231.300	15-Apr-2024	13412

Figure 3 Published survey list

4 Create Survey

Steps to Create Survey: -

- Launch the BQS Application as Admin.
- Click on Create Survey under Survey.
- In create survey there are three headers, those are:
 - **Survey Details**
 - **Surveyor and Launch Service Taken by**
 - **Nominated Fuel Grade**

4.1 Survey Details:

- Enter all the Survey Details. In this we find the following fields are:
- Client, Client Nationality, Client Party, Vessel IMO and its Vessel Name, Port UN/LOCODE, Port, Location, Survey Date, CE Name, ID, Nationality, Captain Name, ID, Nationality, Voyage number will be entered/selected manually as shown in the following screen.

4.2 Surveyor and Launch Service Taken By:

- Under this Details of the surveyor who carry out the surveying process will be available.
- It also contains the details of Bunker Surveyor License Number and Bunker Surveyor Name [fetched automatically], Boat Arranged By: Pier → Vessel and Vessel → Pier, Job Number and Launch Service Taken [selected manually].

4.3 Nominated Fuel Grade:

- In this section we find the following fields which needs to be filled, the fields are:
- Fuel Grade, Nominated Qty, Supplier Density, Bunker Broker Name, ID, Physical Supplier Name, ID, Bunker Tanker IMO, SB Number, Bunker Tanker Name, Bunker Tanker Rep Name, ID, Delivery type (It can be either MFM or Manual Gauge).

The screenshot displays the 'CREATE SURVEY' interface within the POOL BSOL application. The form is organized into three primary sections:

- SURVEY DETAILS:** Includes fields for Client, Client Nationality, Client Party, IMO, Vessel, Port UN/LOCODE, Port, Location, Survey Date, CE Name, CE ID, CE Nationality, Captain Name, Captain ID, Captain Nationality, and Voyage.
- SURVEYOR AND LAUNCH SERVICE TAKEN BY:** Includes fields for ID/Licence Number, Name, Boat Arranged By (Pier → Vessel), Boat Arranged By (Vessel → Pier), and Job Number.
- NOMINATED FUEL GRADE:** Includes fields for Grade, Nominated Qty, Supplier Density, Bunker Broker Name, Bunker Broker ID, Physical Supplier Name, Physical Supplier ID, Bunker Tanker IMO, SB Number, Bunker Tanker Name, Bunker Tanker Rep Name, and Bunker Tanker Rep ID.

At the bottom of the form, there is an 'Add Grade' button, a 'Cancel' button, and a 'Create Survey' button. A footer note states: 'ID * - Identification No/Licence No'.

Figure 4

Figure Create survey

- Click on Create Survey Button.
- Survey will be created in Web Application. Once the Survey is created, it will be navigated to Summary Screen.

TTTT / / 24-May-24 - Open

POOL BSOL

^ SUMMARY

SURVEY DETAILS

Client Client Nationality Client Party IMO Vessel Port UN/LOCODE

Port Location Survey Date CE Name CE ID CE Nationality

Captain Name Captain ID Captain Nationality Voyage No

SURVEYOR AND LAUNCH SERVICE TAKEN BY

ID/Licence Number Name Boat Arranged By:Pier → Vessel Boat Arranged By:Vessel→Pier Job Number

LSMGO savita karagi OWNER SURVEYOR

NOMINATED FUEL GRADE

Grade Nominated Qty Supplier Density Bunker Broker Name Bunker Broker ID Physical Supplier Name

Physical Supplier ID Bunker Tanker IMO SB Number Bunker Tanker Name Bunker Tanker Rep Name Bunker Tanker Rep ID

Delivery Type

☐ Note of Protest

ID * - Identification No/Licence No

^ SURVEY INFO - LSMGO

Figure 5 Summary Screen

Note:

- By clicking on add grade button surveyor can add multiple fuel grades.
- Maximum 3 grades are allowed to select under Nominated fuel grade.
- Master data can be created/updated while creating the Survey using the pencil icon present against the fields.

5 Summary Details:

- This screen is displayed, after completing creation of survey and it contains 3 tabs like: -
 - **Summary**
 - **Survey Info**
 - **Remarks**



Figure 6 Summary Tabs

5.1 Summary:

- Under **Summary** tab there are three Headers, those are as follows: -
 - a) Survey Details
 - b) Surveyor and Launch Service taken by
 - c) Nominated Fuel Grade

Note:

- For above tabs, details entered in create survey screen will be fetched as shown in below screen.

TTTT / 24-May-24 - Open

POOL BSOL

BSOL

↑ SUMMARY

SURVEY DETAILS

Client: [] Client Nationality: [] Client Party: [] IMO: 6543443 Vessel: TTTT Port UN/LOCODE: []

Port: [] Location: [] Survey Date: 11-JUN-2024 CE Name: [] CE ID: [] CE Nationality: []

Captain Name: [] Captain ID: [] Captain Nationality: [] Voyage No: []

SURVEYOR AND LAUNCH SERVICE TAKEN BY

ID/Licence Number: savita203 Name: savita karagi Boat Arranged By:Pier → Vessel: OWNER Boat Arranged By:Vessel→Pier: SURVEYOR Job Number: []

NOMINATED FUEL GRADE

Grade: LSMGO Nominated Qty: [] Supplier Density: [] Bunker Broker Name: [] Bunker Broker ID: [] Physical Supplier Name: []

Physical Supplier ID: [] Bunker Tanker IMO: [] SB Number: [] Bunker Tanker Name: [] Bunker Tanker Rep Name: [] Bunker Tanker Rep ID: []

Delivery Type: PHYSICAL GUAGE Note of Protest: ☐

ID * - Identification No/Licence No

✓ SURVEY INFO - LSMGO

Figure 7 Summary details.

5.2 Survey Info:

- Under **Survey Info** tab there are 2 Headers and those are:
 - a) **Bunker Delivery Summary:** - Under this header there are two tables and those are:
 - **Initial ROB:** - Initial ROB have 3 fields as Actual, Logbook, & Difference. All fields fetch the data from **ROB** screen.
 - **Vessel and Barge Management:** - Under this table **BDN Quantity** field is manual entry. **Vessel Received Quantity** fetches the data from **VMR** Screen and **Surveyor Barge Quantity** fetches the data from either **BTMR** or **MFM** screen based on selection of Delivery Type.

↑ SURVEY INFO - LSMGO

BUNKER DELIVERY SUMMARY

Initial ROB		
Actual	Log-Book	Difference

Vessel And Barge Measurement						
Grade	(A) BDN Qty(Mt)	(B) VSL Received Qty (Mt)	(C) Surveyor Barge Qty (Mt)	B-A	B-C	Diff From Nominated Qty(Mt)
LSMGO						

Figure 8 ROB-Bunker Delivery Summary

- (b) **Time-log:** - All the Time log events with Date and Time is displayed. This data will be fetched from **Time-log** Screen, and it is non-editable. The total number of hours for survey is shown at the end. As Shown in the following screen.

TIMELOG						
Event	Date	Time	Delay Hours	Reason	Comments	
VESSEL ARRIVED (EOSP)						
PILOT ONBOARD (POB)						
SURVEYOR LEFT BASE						
SURVEYOR ARRIVED PIER / BERTH						
VESSEL ANCHORED/BERTHED						
SURVEYOR DEPARTED PIER (FOR ANCHORAGE)						
SURVEYOR ARRIVED ONBOARD						
BUNKER TANKER ALONGSIDE						
SURVEYOR BOARDED BUNKER TANKER						
HOSE CONNECTED						
COMMENCE PUMPING						
COMPLETED PUMPING						
DOCUMENTATION COMPLETED						
HOSE DISCONNECTED						
BUNKER TANKER CAST-OFF						
SURVEYOR LEFT VESSEL						
SURVEYOR ARRIVED PIER						
SURVEYOR RETURNED BASE						
TOTAL SURVEY HR (ACCTS)						

Figure 9 Summary-Time Log

5.3 Remarks:

- Any remarks during the bunkering process are written in this field.

^ REMARKS

X Cancel
Submit Survey

Figure 10 Summary-Remarks

Points to Remember:

- If the Delivery type is **MF**M, then Surveyor Barge Quantity fetches the data from MFM. If the Delivery type is **Manual Gauge**, it fetches the data from BTMR screen.
- If there is more discrepancy found during bunkering process, surveyor will tick **Note of Protest** checkbox.

6 Pre-Survey Ship's Acknowledgement:

This process is done before bunkering where C/E will take samples from barge. There are 2 tabs **Pre-survey** and **Remarks**,

Under Pre-survey tab, there 3 headers and those are **Sampling**, **General Information**, **Caution (signing of document and internal transfer of bunkers)** and **Safety**.

1. Pre-Survey: -

- Sampling
- General Information
- Caution (signing of document and internal transfer of bunkers) and Safety.

2. **(a). Sampling:** - Here, the bunker surveyor will be supervising the collection of the representative sample during bunkering operations like number of samples taken, laboratory name, location (it is drop down with options to select Ship's manifold and Barge's manifold). As shown below

PRE-SURVEY SHIP'S ACKNOWLEDGEMENT

We have been appointed by Messrs. **Client** to conduct a bunker quantity survey of the bunker to be delivered to the ship. Please note that the attending surveyor from **Pool BSOL Ltd.** will be carrying out the sur... [Read more](#)

SAMPLING

The attending bunker surveyor will be witnessing and supervising the collection of the representative sample during the bunkering operations. The sampling shall be carried out on the ship's bunker manifold as per the requirements set out in Singapore Standard SS648 Code & SS648 of Practice for Bunkering (latest edition). The samples collected will be subsequently sealed, labelled and distributed accordingly.

Will you, Chief Engineer, be taking your own sample(s) ? If Yes, number of samples Location

Will you be sending your own sample(s) for analysis ? If Yes, to which laboratory

Figure Presurvey ships acknowledgement

- (b). General Information:** - In this table, grade and quantity is fetched from Nominated fuel grade. Bunkering tank Sequence and Required Pumping should be entered manually as shown below.

GENERAL INFORMATION						
SHIP's Bunker Nomination	Grade	Quantity	Grade	Quantity	Grade	Quantity
	LSMGO					
SHIP's Bunkering Tanks Sequence						
SHIP's Requested Manifold Pressure						
SHIP's Requested Pumping Rate						

Figure General information

- (c). Caution (signing of document and internal transfer of bunkers) and Safety:** - It consists of drop down Yes/No for the options present under this header, as shown below.

CAUTION (SIGNING OF DOCUMENT AND INTERNAL TRANSFER OF BUNKERS)

Was Chief Engineer advised not to pre-sign any documents related to this bunkering until the job has been completed and to consult surveyor when in doubt ?

YES ▼

Was Chief Engineer advised not to carry out any internal transfer during the bunkering operation ?

NO ▼

SAFETY

Will ship provide SOLAS approved safe access to bunker tanker for Surveyor ?

YES ▼

Remarks: -

Any remarks during the bunkering process are written in this field.

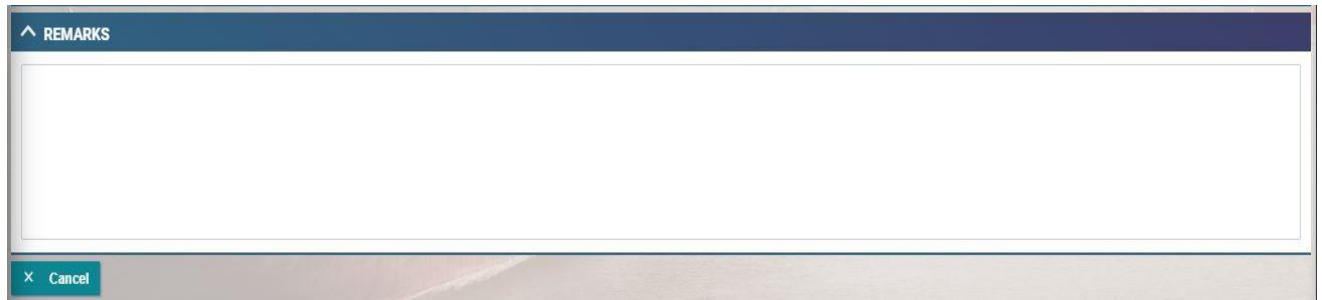
A screenshot of a software interface showing a large, empty text area for entering remarks. The text area is framed by a dark blue header bar with the word "REMARKS" and a small upward-pointing arrow. Below the text area is a green button with a white "X" icon and the word "Cancel".

Figure presurvey11 Remarks

7 Time log:

- A time log is a chronological set of records that document of the sequence of bunkering activities or planned bunkering activities. The time log provides a step-by-step record by which time-related data can be traced to their source and provides a complete history of all bunkering activities with a date and time.
- There are two tabs in Time log screen, and they are: -
 - Time Log.
 - Remarks.

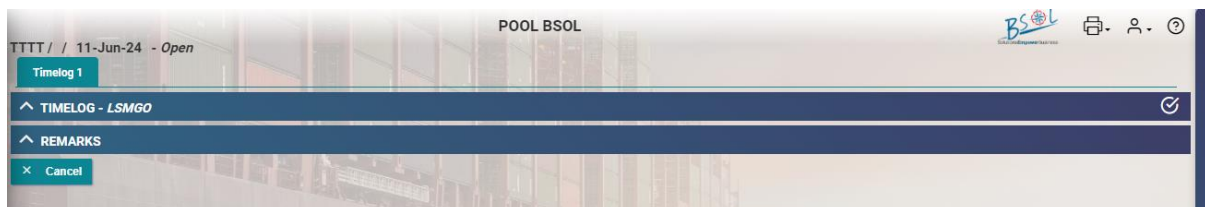
A screenshot of a software interface showing two tabs. The top tab is labeled "TIMELOG - LSMGO" and has a green checkmark icon. The bottom tab is labeled "REMARKS" and has a green "X" icon and the word "Cancel". The background of the interface shows a blurred image of a building.

Figure 12 Time Log Tabs

7.1 Time log details:

- In this screen, the Time-log events are listed. The bunker surveyor shall record the time of all relevant events related to the entire bunkering operation until their return to base.
- Date and Time need to be entered for each Time-Log Event, the date format should be DD-MMM-YYYY & time format should be HHMM.

TTTT / 11-Jun-24 - Open

POOL BSOL

BSOL

TimeLog 1

TIMELOG - LSMGO

Event	Date	Time	Delay Hours	Reason	Comments
VESSEL ARRIVED (EOSP)	DD-MMM-YYYY				
PILOT ONBOARD (POB)	DD-MMM-YYYY				
SURVEYOR LEFT BASE	DD-MMM-YYYY				
SURVEYOR ARRIVED PIER / BERTH	DD-MMM-YYYY				
VESSEL ANCHORED/BERTHED	DD-MMM-YYYY				
SURVEYOR DEPARTED PIER (FOR ANCHORAGE)	DD-MMM-YYYY				
SURVEYOR ARRIVED ONBOARD	DD-MMM-YYYY				
BUNKER TANKER ALONGSIDE	DD-MMM-YYYY				
SURVEYOR BOARDED BUNKER TANKER	DD-MMM-YYYY				
HOSE CONNECTED	DD-MMM-YYYY				
COMMENCE PUMPING	DD-MMM-YYYY				
COMPLETED PUMPING	DD-MMM-YYYY				
DOCUMENTATION COMPLETED	DD-MMM-YYYY				
HOSE DISCONNECTED	DD-MMM-YYYY				
BUNKER TANKER CAST-OFF	DD-MMM-YYYY				
SURVEYOR LEFT VESSEL	DD-MMM-YYYY				
SURVEYOR ARRIVED PIER	DD-MMM-YYYY				
SURVEYOR RETURNED BASE	DD-MMM-YYYY				

TOTAL SURVEY HR (ACCTS)

BUNKER TANKER BERTHING DELAY

Figure 13 Time Log Details

7.2 Remarks:

- Any remarks during the bunkering process are written in this field.

REMARKS

Cancel

Figure 14 Time Log - Remarks

8 ROB (Remaining On-Board):

- It includes details of the material remaining on vessel tanks, void spaces after discharge. Remaining On-Board Quantity includes water, oil, oil residue, oil/water emulsions, sludge, and sediment.
- There are 3 tabs, and they are as follows: -
 - **Bunker R.O.B quantity report**
 - **Hands-On table**
 - **Remarks**

Figure 15 ROB

8.1 Bunker R.O.B quantity report:

- The cargo officer shall declare the arrival remaining on board as per the stock movement logbook and invite the chief engineer to witness the opening gauge (meter reading) and the taking of cargo temperature of all cargo tanks of the bunker tanker.
- Under this tab we have Calibration tables/chart by, calibration date (DD-MMM-YYYY), Chart/table status (selected manually), Surveyor sounding tape, Multigas Detector and Thermometer.
- Draft Fwd. and Draft Aft, Trim [it's in disabled state and it's the difference between draft aft & draft fwd. (draft aft - draft fwd.)], Trim correction applied.
- **Tilt Direction** field (it consists of dropdown Starboard, and Port need to be selected) and it is enabled by giving the value in **List** field. List Correction applied is dependent on List field.
- **Sea State** field (it consists of dropdown Calm, Slight, Moderate and Rough need to be selected) based on the sea behaviour.
- **Default Measurement** field (it consists of dropdown Sounding or Ullage need to be selected) based on the measurement instrument used during measuring the tanks.
- In this Report, all the data needs to be entered manually.

Figure 16 ROB-Bunker ROB Quantity Report

8.2 Hands on Table:

- By default, Nominated Grade (The grade which is entered while creating Survey) will be displayed here.
- **Logbook** field is manual entry where **Surveyor's ROB** field is total calculated in hands on table, Difference (it's the difference between Surveyor's ROB and Logbook).
- Tanks can be categorized into Nominated and Non-nominated Tanks. Need to select nominated Tanks (through check box) to which the purchased bunkers will be transferred. This will automatically get reflected in VMR Screen.
- New grade can be added by clicking on "**Grade +**",

- The grade which is going to add in ROB screen that is Non-Nominated Grade. which creates one more tank table.

LSMGO		LOG BOOK			SURVEYOR'S ROB			DIFFERENCE								
Nom.	Tank	Temp (°C)	100% Capacity	Ref. Height (m)		Obsd. SDG/ULL	Corrd. SDG/ULL	Total Obsd. Vol. (m³)	Free Water		Gross Obsd. Vol. (m³)	Density @ 15°C	VCF	Gross Std. Vol. (m³)	WCF	Net Quantity (Mt)
			Vol (m³)	Table	Measured				SDG	Vol (m³)						
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Figure 17 ROB-Hands on Table.

8.3 Remarks:

- Any remarks during the bunkering process are written in this field.

^

REMARKS

X

Cancel

Figure 18 ROB-Remarks

9 VMR (Vessel Measurement Report):

- The data, which is entered in ROB will get reflected in VMR screen.
- It has 4 tabs, and they are as follows: -
 - **Vessel Measurement Report**
 - **Reading Summary**
 - **Opening Reading**
 - **Closing Reading**
 - **Remarks**

The screenshot displays the 'VESSEL MEASUREMENT REPORT - LSMGO' screen. The 'READING SUMMARY' tab is active, showing fields for 'Supplier Density', 'Total Std. Vol. Received', 'W.C.F Table-56', and 'Net Bunker Quantity', all with a value of 0. Below this, there are sections for 'OPENING READING' and 'CLOSING READING', both with a 'TOTAL' field and a 'NO CHANGE' toggle set to 'NO'. A 'REMARKS' section is also visible at the bottom.

Figure 19 VMR

9.1 Vessel Measurement Report:

- In this tab, we find the Calibration tables/charts by, Calibration date, (fetches data from ROB) Chart /Table status, (manual selection), Surveyor Sounding Tape, Multigas Detector and Thermometer (fetches data from the ROB).

The screenshot displays the 'VESSEL MEASUREMENT REPORT - LSMGO' screen. The 'Calibration Tables/Chart By' tab is active, showing fields for 'Calibration Date' (with a date picker), 'Surveyor Sounding Tape', 'MultiGas Detector', and 'Thermometer'.

Figure 20 VMR-Vessel Measurement Report

9.2 Reading Summary:

- This consists of Supplier Density(it fetches data from the summary screen), Total Std Volume Received, WCF Table, and Net Bunker Quantity. These 3 fields are in disable state and data comes based on calculation done on the opening and closing reading tabs.

The screenshot displays the 'VESSEL MEASUREMENT REPORT - LSMGO' screen. The 'READING SUMMARY' tab is active, showing fields for 'Supplier Density', 'Total Std. Vol. Received', 'W.C.F Table-56', and 'Net Bunker Quantity', all with a value of 0.

Figure 21 VMR-Reading Summary

Note:

- Total Std Volume Received is calculated by using the formula “Closing Reading Total Gross Std Vol - Opening Reading Total Gross Std Vol”.

- It consists of Draft fwd. and Draft aft field, Trim field, Trim correction applied, List, List correction applied, Sea state and Default measurement fields.
- In Closing Reading section, Tank, Capacity and Reference height will be reflected from ROB, remaining data need to be entered manually.
- If “**No Change**” is enabled (Yes), then non-Nominated tank details will be copied from Opening Reading Section. But, for Nominated tank, data needs to be entered.
- The Sounding/Ullage can be specified by typing S or U at the end of the figure. By default, this will be sounding. If it is a Gauge, then type G in the input field. The table as Show in the following screen.

[illegible]

Figure VMR-Closing Reading

Note: -

- **Sounding** is the measured height of the fluid from the surface of the fluid to the bottom of the tank.
- **Ullage** is the void space in the tank measured from the top of the tank to the upper surface of the fluid.

9.5 Remarks:

- Any remarks during the bunkering process are written in this field.

^

REMARKS

X

Cancel

Figure VMR-Remarks

10 BMR (Bunker Tanker (Barge) Measurement Report):

- The form is used to record tank gauging and cargo temperature readings of bunker tanker as witnessed by the chief engineer and bunker surveyor.
- This report is used by the cargo officer of the bunker tanker to record the calculation of the quantity in the tanks.
- It has the following tabs, and they are: -
 - **Barge Measurement Report**
 - **Reading Summary**
 - **Opening Readings**
 - **Closing Readings**
 - **Remarks.**

TTT / / 11-Jun-24 - Open POOL BSOL

BARGE MEASUREMENT REPORT - LSMGO

BARGE SAMPLE INFO

Drip Sampling container Seal No: Needle Valve Seal No: Sufficient Tank Space for line-cleaning:

READING SUMMARY

Supplier Density: Total Std. Vol. Received: W.C.F Table-56: Bunker Qty Delivered:

OPENING READING LSMGO: TOTAL:

CLOSING READING LSMGO: TOTAL: NO CHANGE: ☐ NO: ☐

BARGE MEASUREMENT WITNESS

Barge opening measurements witnessed by ship representative, if yes, Designation:

Barge closing measurements witnessed by ship representative, if yes, Designation:

REMARKS

Figure BTMR

10.1 Barge Measurement Report:

- In this tab, we find the Calibration tables/charts by (manual entry), Calibration date (Date format should be DD-MM-YYYY), Chart/Table status (its drop down and have the following options true, duplicate, Loose) and Surveyor sounding tape, Multigas Detector and Thermometer.

TTT / / 11-Jun-24 - Open POOL BSOL

BARGE MEASUREMENT REPORT - LSMGO

BARGE SAMPLE INFO

Drip Sampling container Seal No: Needle Valve Seal No: Sufficient Tank Space for line-cleaning:

READING SUMMARY

Supplier Density: Total Std. Vol. Received: W.C.F Table-56: Bunker Qty Delivered:

Figure BTMR-Bunker Tankers Measurement Report

10.2 Reading Summary:

- Under this header we find the following fields Supplier Density, Total Gross Std. Volume, WCF Table-56, Bunker Quantity Delivered as Shown in the following screen.

READING SUMMARY

Supplier Density: Total Std. Vol. Received: W.C.F Table-56: Bunker Qty Delivered:

Figure BTMR-Reading Summary

Note:

- Total Std vol Delivered is calculated by using the formula “Opening Reading Total Std Vol- Closing Reading Total Std Vol”.
- WCF Table -> Supplier Density -0.0011
- Bunker Quantity Delivered is calculated by using the “Opening Reading Total Net Qty- Closing Reading Total Net Qty”.
- Here the total calculation will happen only for the Nominated Tanks.

10.3 Opening Reading Table:

- The Measurements of the **Bunker Tanker (Barge)** before bunkering will be recorded under **Opening Reading Section**.
- This is a hands-on table and in this table we record the measurements of the barge.
- It consists of Draft fwd. and Draft aft field, Trim field, Trim correction applied, List, List correction applied, Sea state and Default measurement fields, Sea state based on the sea behaviour and Default measurement fields, based on the measurement instrument used during measuring the tanks, all the data in these fields is entered manually and the table is shown in the following screen.

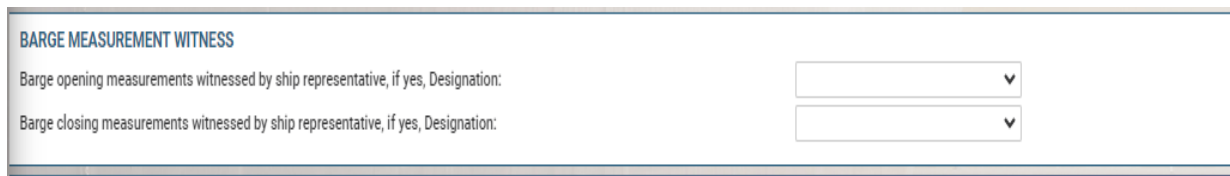
Figure BTMR-Opening Reading

10.4 Closing Reading Table:

- The measurements of the **Bunker Tanker (Barge)** after bunkering will be recorded under **Closing Reading Section**.
- In Closing Reading section, Tank, Capacity and Reference height will be reflected from Opening reading, remaining data need to be entered manually.
- If “**No Change**” is enabled (Yes), then tank details will be copied from Opening Reading Section. If any change is required, then it needs to alter.

Figure BTMR -Closing Reading

10.5 BARGE MEASUREMENT WITNESS:



BARGE MEASUREMENT WITNESS

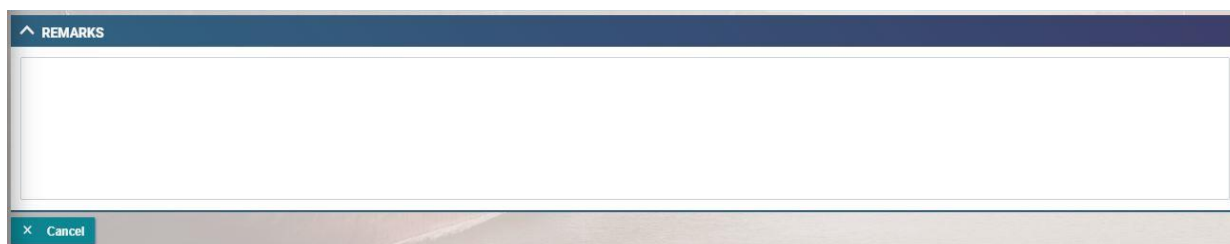
Barge opening measurements witnessed by ship representative, if yes, Designation:

Barge closing measurements witnessed by ship representative, if yes, Designation:

Figure BTMR-Barge Measurement witness

10.6 Remarks:

- Any remarks during the bunkering process are written in this field.



REMARKS

Cancel

Figure BTMR-Remarks

- If the sampling requirements provided by the bunker tanker differs, the chief engineer shall make his comments in the “Remark” part of the bunker tanker measurement report. The bunker surveyor shall log this fact in the statement of facts of the bunker survey report.

11 MFM (Mass Flow Meter):

- Mass Flow Metering is a new technology that will enable bunkering operations to be more transparent. Flowmeter is an instrument for measuring the rate of flow of water, gas, or fuel, especially through a pipe.
- It has the following tabs and they are: -
 - (a) Before Delivery
 - (b) After Delivery
 - (c) Reading Summary
 - (d) Barge Sample Info
 - (e) Barge Reading Witness
 - (f) Remarks

▼ BEFORE DELIVERY	
▼ AFTER DELIVERY	
READING SUMMARY	
(B-A) - (Y-X) Quantity Delivered	Bunker Metering Ticket (BMT)
0	
BARGE SAMPLE INFO	
Drip Sampling container Seal No	Needle Valve Seal No
Suff. Tank Space for line-cleaning	
BARGE READING WITNESS	
Barge opening readings witnessed by ship representative, if yes,	Select
Designation:	NA
Barge closing readings witnessed by ship representative, if yes,	Select
Designation:	NA
▼ REMARKS	

Figure MFM

(a). Before Delivery: -

- This screen's purpose is to note down the Before Bunkering process and also types of flowmeter (like Endress Hauser, Emerson etc.) used in bunkering process along with bunker density, temperature, its mass(forward mass and reverse mass inventory), supply tanks(to which tank of the vessel bunker(fuel) is required). As shown below

^ BEFORE DELIVERY				
Type Of Flowmeter	(A) Forward Mass Inventory Reading	(X) Reverse Mass Inventory Reading	Temperature	Density @ 15°C
Supply Tanks	Seal Number Issued by Weights and Measures Office			

Figure MFM-Before Delivery

(b). After Delivery: -

- This screen's purpose is to note down the After-bunkering process and mass inventory used in bunkering process along with bunker density, temperature, supply tanks (to which tank of the vessel bunker(fuel) is required). As shown below

^ AFTER DELIVERY				
(B) Forward Mass Inventory Reading	(Y) Reverse Mass Inventory Reading	Temperature	Density @ 15°C	Supply Tanks

Figure MFM-After Delivery

(c). Reading Summary: -

- Based on the Before and After Delivery (forward mass inventory and reverse mass inventory readings), the **Quantity Delivered** is calculated in this header.
- Bunker Metering Ticket (BMT) field is provided.

READING SUMMARY	
(B-A) - (Y-X) Quantity Delivered	Bunker Metering Ticket (BMT)
0	

Figure MFM-Reading Summary

(d). Barge Sample Info: -

- Barge sample info is also taken into consideration in which Drip Sampling container seal no., Needle valve seal no., Sufficient Tank Space for line-cleaning (dropdown Yes/No is provided).

BARGE SAMPLE INFO		
Drip Sampling container Seal No	Needle Valve Seal No	Suff. Tank Space for line-cleaning

Figure MFM-Barge Sample Info

(e). Barge Reading witness: -

- Barge opening and closing readings are noted by ship representatives in the presence of Chief engineer, 1st, 2nd, 3rd, 4th engineer, cadet and other list of options provided in drop down of Barge Reading Witness tab.

BARGE READING WITNESS				
Barge opening readings witnessed by ship representative, if yes,	Select	NA	NA	NA
Designation:				
Barge closing readings witnessed by ship representative, if yes,	Select	NA	NA	NA
Designation:				

Figure MFM-Barge Reading Witness

(f). Remarks: -

- Remarks tab is present in the screen if there are any discrepancies or disputes during bunkering can be entered in this field.

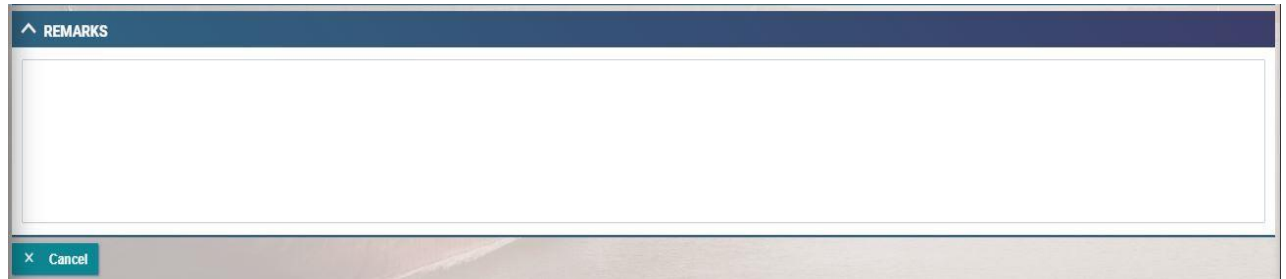
A screenshot of a software interface showing a 'REMARKS' tab. The tab is highlighted in dark blue with a white upward-pointing arrow and the word 'REMARKS' in white. Below the tab is a large, empty white rectangular area for text entry. At the bottom left of the tab area is a small green button with a white 'x' icon and the word 'Cancel' in white.

Figure MFM-Remarks

12 Sampling Form:

- The taking of the Primary Sample is to be at the Point of Custody Transfer at the vessel's manifold.
- It has 2 tabs, and they are as follows: -
 - **Sampling Information.**
 - **Record Of Sub-Divided Samples, Seals, Counter- Seals and Distribution Record.**

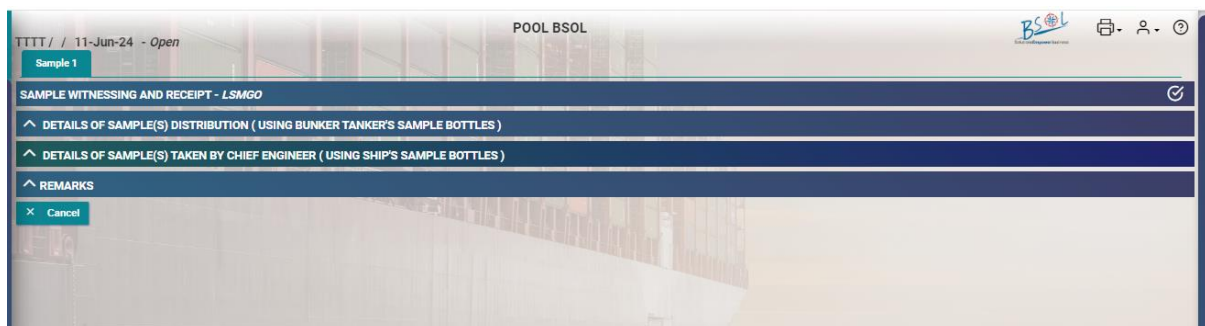
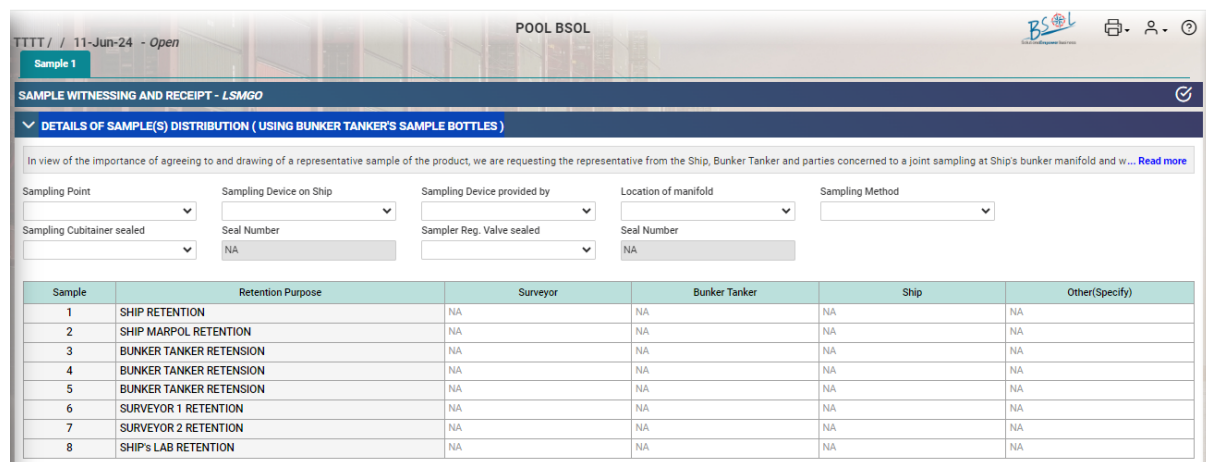
A screenshot of a software interface for 'Sample 1'. The top bar shows 'TTTT / 11-Jun-24 - Open' on the left, 'POOL BSOL' in the center, and a logo with 'BSOL' on the right. Below the top bar is a list of tabs: 'SAMPLE WITNESSING AND RECEIPT - LSMGO', 'DETAILS OF SAMPLE(S) DISTRIBUTION (USING BUNKER TANKER'S SAMPLE BOTTLES)', 'DETAILS OF SAMPLE(S) TAKEN BY CHIEF ENGINEER (USING SHIP'S SAMPLE BOTTLES)', and 'REMARKS'. The 'REMARKS' tab is currently selected and highlighted in dark blue. At the bottom left of the tab area is a small green button with a white 'x' icon and the word 'Cancel' in white.

Figure Sampling Information

12.1 Sampling Information:

- The sampling information tab contains the fields as shown in the following screen.



Sample	Retention Purpose	Surveyor	Bunker Tanker	Ship	Other(Specify)
1	SHIP RETENTION	NA	NA	NA	NA
2	SHIP MARPOL RETENTION	NA	NA	NA	NA
3	BUNKER TANKER RETENSION	NA	NA	NA	NA
4	BUNKER TANKER RETENSION	NA	NA	NA	NA
5	BUNKER TANKER RETENSION	NA	NA	NA	NA
6	SURVEYOR 1 RETENTION	NA	NA	NA	NA
7	SURVEYOR 2 RETENTION	NA	NA	NA	NA
8	SHIP'S LAB RETENTION	NA	NA	NA	NA

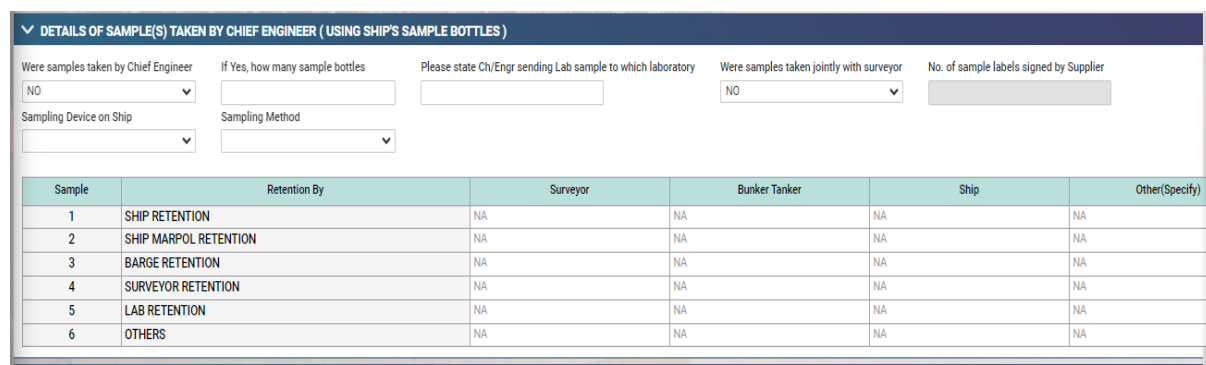
Figure 22 Sampling Information- details of sample(s) distribution (using bunker tanker's sample bottles)

Note:

- Under Sampling information tab most of the field will be set to default values, and user will have the access to modify the data.

12.2 Record Of Sub-Divided Samples, Seals, Counter- Seals and Distribution Record:

- This tab depicts the field like “The Primary Sample was divided into how many bottles?” And will be accept the numbers.
- Also, this tab has a table to record the details of Seal Number, Supplier's Seal Number and Surveyor's Seal Number (If appointed) for each of Sample Distribution listed as Show in the following screen.



Sample	Retention By	Surveyor	Bunker Tanker	Ship	Other(Specify)
1	SHIP RETENTION	NA	NA	NA	NA
2	SHIP MARPOL RETENTION	NA	NA	NA	NA
3	BARGE RETENTION	NA	NA	NA	NA
4	SURVEYOR RETENTION	NA	NA	NA	NA
5	LAB RETENTION	NA	NA	NA	NA
6	OTHERS	NA	NA	NA	NA

Figure 23 sampling information- details of sample(s) taken by chief engineer (using ship's sample bottles)

12.3 Remarks: -

- Remarks tab is present in the screen if there are any discrepancies or disputes during bunkering can be entered in this field.

Figure 24 Sampling Information -Remarks

13 Barge-Nc (non-cargo declaration):

- The non-cargo tank declaration/inspection form is a declaration by the cargo officer on the gauging and contents of all the non-cargo tanks and spaces as in the General Arrangement (GA) plan of the bunker tanker.
- It has following tabs and they are as follows: -
 - **Before Bunkering**
 - **After Bunkering**
 - **Remarks**

Figure 25 BARGE NC

13.1 Before Bunkering:

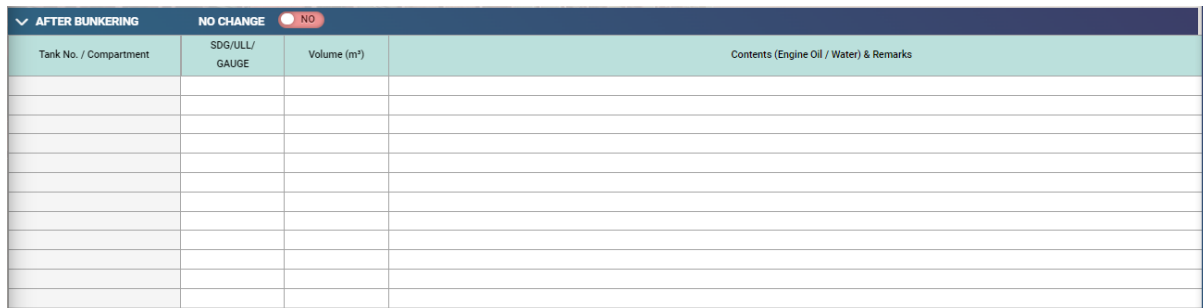
- In this, we have table for “Before Bunkering” in which we enter the data for each tank and content with measurement Gauge and volume by giving Remarks of what is present in the barge during bunkering operation.
- The table is Show in the following screen.

Tank No. / Compartment	SDG/ULL/ GAUGE	Volume (m³)	Contents (Engine Oil / Water) & Remarks

Figure 26 Barge NC-Before Bunkering

13.2 After Bunkering:

- There is “No Change” switch in After bunkering table which copies the data from Before bunkering when it is turned to “YES” if there is no difference between after and before bunkering.
- The table is shown in the following screen.



Tank No. / Compartment	SDG/ULL/ GAUGE	Volume (m³)	Contents (Engine Oil / Water) & Remarks

Figure 27 Barge NC-After Bunkering

- **Tank No / Compartment:** Name of non-cargo tanks will be entered here.
- **Sounding (SDG)** is the measured height of the fluid from the surface of the fluid to the bottom of the tank.
- **Ullage (ULL)** is the void space in the tank measured from the top of the tank to the upper surface of the fluid.

13.3 Remarks:

- Any remarks during the bunkering process are written in this field.

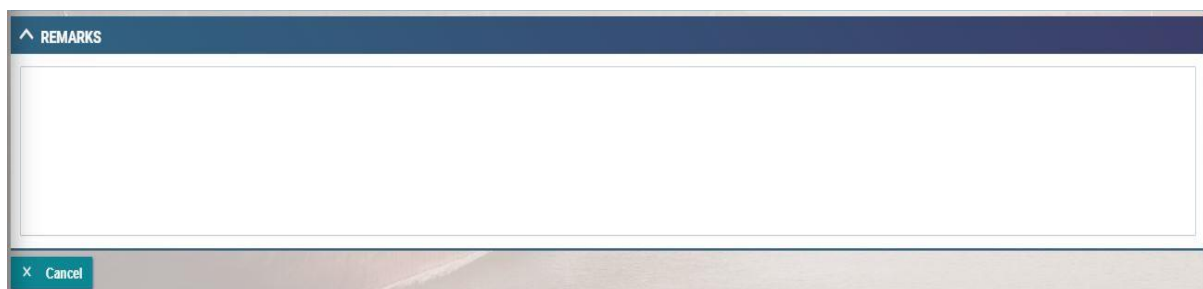


Figure 28 Barge NC -- Remarks

14 SOF (Statement of Fact):

- Any dispute occurred such as Quantity of fuel supplied, Quality of fuel, any malfunctions of the gauges etc. while bunkering will be noted in this screen.
- It has following tabs and they are: -
 - **SOF Summary**
 - **Statement of fact**



Figure 29 SOF

14.1 SOF Summary:

In this screen, all the data is un-editable and those are: -

- “Nomination Qty” [Data is fetching from Summary screen from Nominated Qty field]
- “Bunker Delivery Note (BDN)” [Data is fetching from Vessel and Barge measurement table “(A) BDN Qty (Mt)” under Survey Info tab in Summary screen.]
- “Vessel Received Qty” [Data is fetching from Vessel received Qty (Mt) field in Survey Info tab in Summary screen under section Vessel and Barge measurement table “(B)”]
- “Barge Supplied Qty” [Data is fetching from Surveyor Barge Qty (Mt) field in Survey Info tab in Summary screen under section Vessel and Barge measurement table “(C)”]
- “Nominated Qty Difference” field is calculated by the difference of “Bunker Delivery Note” and “Nomination Qty” field.

i.e., Nominated Qty Difference = Bunker Delivery Note - Nominated Quantity.

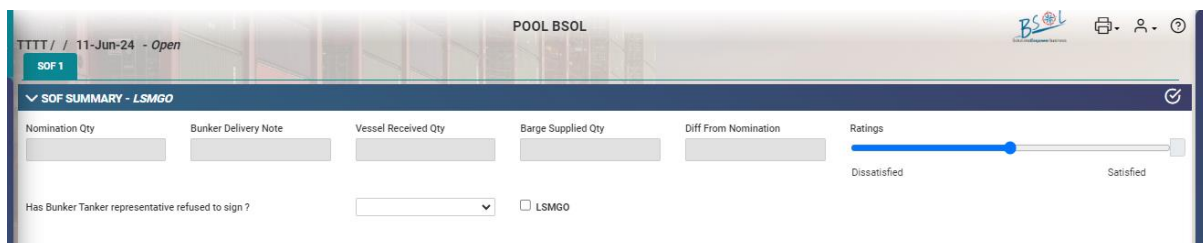


Figure 30 SOF Summary

14.2 Statement of Fact:

- Under this tab the dispute happened while bunkering is noted down.
- User can either type the remarks manually or select the remarks from the dropdown.

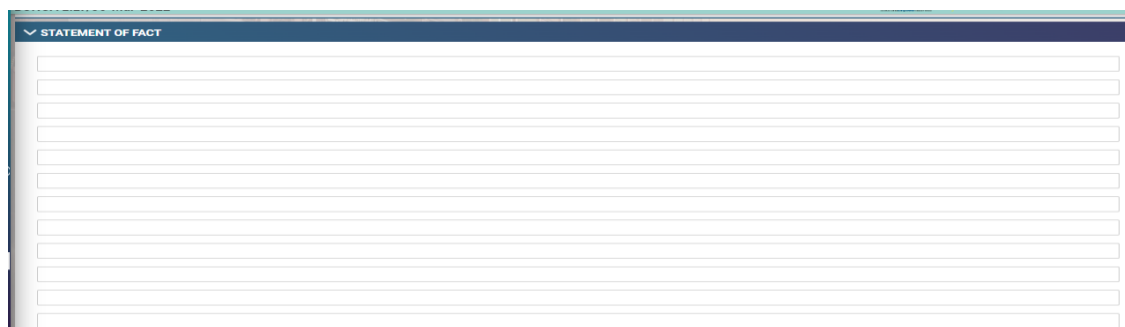


Figure 31 SOF-Statement of Fact

15 Upload File:

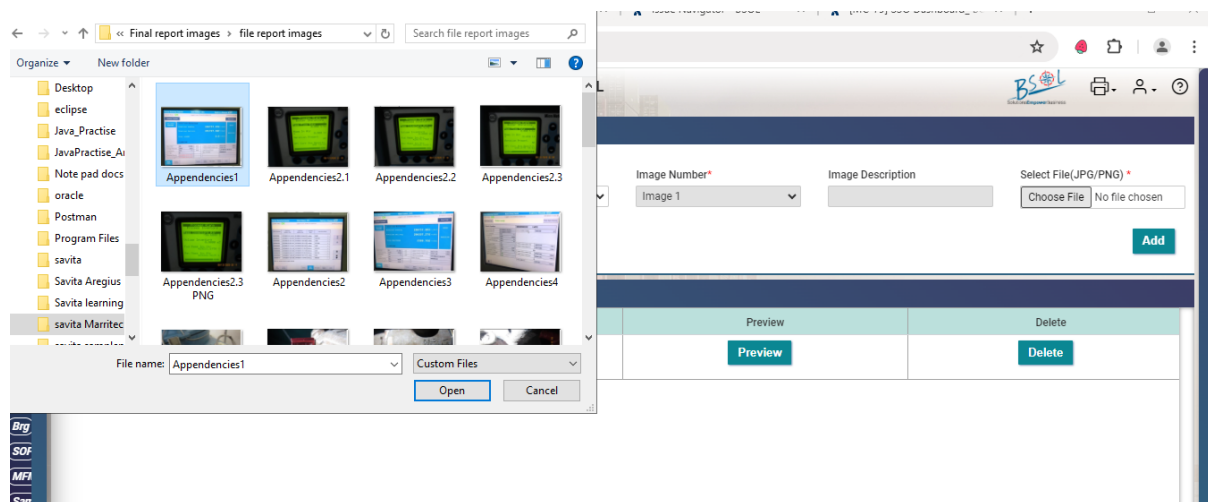
- The purpose of this screen is to upload Document/file by selecting the corresponding file type from the drop-down which user is uploading.
- It has 2 following tabs as Show in the following screen: -
 - Document/File Upload
 - File List



Figure 32 Upload File

15.1 Document/File Upload:

- In this section we have following fields:
- File Type (images), Grade (disable state), Image Number (disable state), Image Text (disable state), Select File (JPG/PNG) here there is option to choose file, by clicking on **Choose file** which opens pop-up and select file and click on open as shown in the following screen.



- In above screen by selecting file, click on Open, As Shown in the following screen.

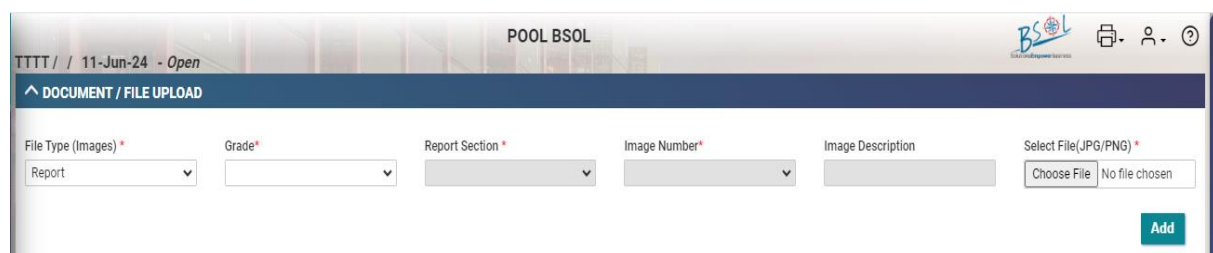


Figure 33 File Upload-Documents/ Upload File

Note:

- Here Based on the selection of File Type(images) (Report) other fields get enabled.

15.2 File List:

- In this section uploaded files will be shown with **File Name and Delete option**.
- User can Preview the uploaded files by clicking on **“Preview”** Button and **“Delete”** Button is provided for deleting the uploaded files.

FILE			
File Name	Image Description	Preview	Delete
photo_present_report_3_13477_1_Grade1_img1.jpg	ff	Preview	Delete
rob_report_3_13477_1_Common_img1.jpg		Preview	Delete
sampling_distribution_report_3_13477_1_Grade1_img1.jpg	ttt	Preview	Delete
sampling_distribution_report_3_13477_1_Grade1_img2.jpg	test2	Preview	Delete
sampling_distribution_report_3_13477_1_Grade1_img3.jpg		Preview	Delete
sampling_information_report_3_13477_1_Grade1_img1.jpg		Preview	Delete

Figure 34 Upload File-File List

16 Note of protest:

- The Note of Protest is a declaration under oath by the Master of the ship.
- It has 2 following tabs as Show in the following screen: -
 - Note of protest
 - Office information

Figure 35 Final Report – MFM Tab details

16.1 Note of Protest:

- This tab contains the Caption Name and Designation dropdown fields and data will be selected manually as shown in the following screen.

Figure 36 Note of Protest details.

- Under the Note of Protest there are five sections, those are as follows: -
 - a) Sampling
 - b) Bunker Quantities

- c) **In Line Blending**
- d) **Cappuccino Bunkers - Excessive Air Blowing**
- e) **Other Reasons (E.g. Undue Delay, Late Arrival, Slow Pumping Etc)**

16.1.1 Sampling

- This tab contains the following checkboxes fields and that are:
 - Supplier refused to witness sampling at point of Custody Transfer at the vessel's manifold,
 - Supplier refused to counterseal vessel's samples which were sub-divided from the primary sample,
 - Supplier refused to allow counterseal of the supplier's samples sub-divided from the primary sample, Supplier refused to record seals/counterseal numbers on to the BDN/BDR

16.1.2 Bunker Quantities

- This tab contains the following checkboxes and data entry fields and that are:
 - The vessel's received quantity was "Quantity", but the BDR / BDN showed "BDR/BDN value" was delivered. We are protesting that the vessel received only "fuel value" of fuel of "Grade Name" Grade.
 - The vessel received insufficient supply of ordered bunker quantity. The Quantity Delivered "Quantity" was of "Grade Name"(Grade) compared with the Quantity Ordered which was "Quantity" of "Grade Name"(Grade).

16.1.3 In Line blending

- This tab contains only text content as:
 - The bunker supplied has been blended in-line during delivery in spite of our objections. This can cause severe machinery damage if fuel is nonhomogeneous. We reserve the right to hold the Supplier responsible for any damages.

16.1.4 Cappuccino Bunkers - Excessive Air Blowing

- This tab contains two date selection fields as Starting date and time, Ending date, and time and both fields can be selected manually.

16.1.5 Other Reasons (Eg Undue Delay, Late Arrival, Slow Pumping Etc)

- This tab contains a field to enter the comments.

16.2 Office Information

- This tab contains the Caption Name and Designation dropdown fields and data will be selected manually.

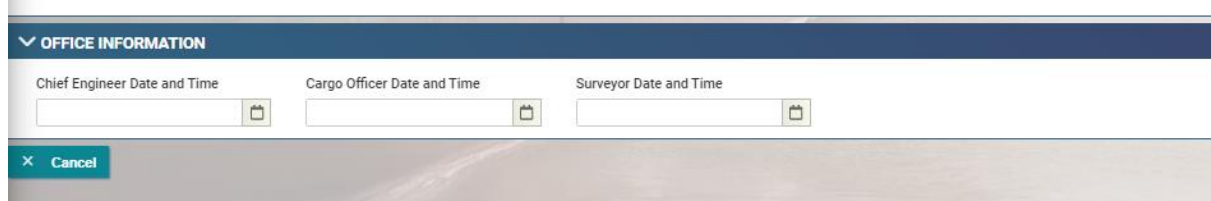


Figure 37 Non MFM -- Office information

17 Pumping Rate

- Under the Pumping Rate there are three sections, those are as follows: -
 - a) **Pumping Rate Record**
 - b) **Summary Of Pumping Rate**
 - c) **Additional Remarks / Comments (If Any)**

The screenshot shows a web application interface for 'POOL BSOL'. At the top, there's a header with 'TTTT / 11-Jun-24 - Open' and a 'PR 1' button. Below this is a section titled 'PUMPING RATE RECORD - LSMGO'. The form is divided into several input fields arranged in a grid. The first row includes 'Vessel' (TTTT), 'IMO No' (6543443), 'Client Name', 'Surveyor Name' (savita karagi), 'Port', and 'Chief Engineer'. The second row includes 'Master', 'Location', 'Date of Attendance' (11-JUN-2024), and an empty field. Below this is a section titled 'SUMMARY OF PUMPING RATE' which contains multiple rows of input fields for 'Bunker Grade' (LSMGO), 'Bunker Line Press. Test (Bar)', 'Bunker Line Press. Test Date', 'Supplier', 'Bunker Tanker Name', 'Bunker Tanker SB No', 'Barge Officer', 'Manifold Pressure Requested', 'Manifold Pressure Achieved', 'BDN Quantity', 'Commenced Pumping', 'Completed Pumping', 'Total Pumping Time', 'Pumping Rate Requested', and 'Pumping Rate Achieved'. At the bottom is a section titled 'ADDITIONAL REMARKS / COMMENTS (IF ANY):' with a dropdown menu labeled 'Please Choose...' and a text area for manual entry.

Figure 38 Pumping Rate

17.1 Pumping Rate Record:

- All the data under this section is fetched from Summary Screen.

17.2 Summary Of Pumping Rate

- In this section, surveyor will enter the data for the fields **Bunker Line Press. Test (Bar)**, **Bunker Line Press. Test Date** and **Manifold Pressure Achieved**.
- Commenced Pumping and Completed Pumping data is fetched from Timelog. The difference between those two records will result in **Total Pumping Time** field.
- **Bunker Grade**, **Supplier**, **Bunker Tanker Name**, **Bunker Tanker SB No**, **Barge Officer** data is fetched from Summary screen.
- **BDN Qty** is fetched from Summary screen under Survey Info section.
- **Manifold Pressure Requested**, **Pumping Rate Requested** is fetched from **Pre-Survey Screen**.
- **Pumping Rate Achieved** is calculated using the formula -> **BDN Qty/Total Pumping Time**.

17.3 Additional Remarks / Comments (If Any)

- User can either type the remarks manually or select the remarks from the dropdown.

18 Vessel Info

- In this screen, Vessel Information will be displayed.
- Vessel Info has three sections, those are as follows: -
 - a) **Vessel Particulars**
 - b) **Bunker Consumptions (Mt Per Day)**
 - c) **Voyage Details**

POOL BSOL

TTTT / 11-Jun-24 - Open

✓ VESSEL PARTICULARS

✓ BUNKER CONSUMPTIONS (MT PER DAY):

✓ VOYAGE DETAILS

✕ Cancel Copy Data To Text

Figure 39 Vessel Info

18.1 Vessel Particulars

- In this section, Vessel Name, IMO Number, LOA, BEAM, Light Ship Weight, Summer DWT, Summer Draft, Vessel Type, Vessel Size, Built Year of the Vessel will be entered.

POOL BSOL

TTTT / 11-Jun-24 - Open

✓ VESSEL PARTICULARS

Vessel Name	IMO No.	LOA	Beam	Light Ship Weight	Summer DWT	Summer Draft
TTTT	6543443	NA	NA	NA	NA	NA
Vessel Type	Vessel Size	M/E Make	M/E Model	M/E Built Year	BHP	
		NA	NA		NA	

Figure 40 Vessel Particulars

18.2 Bunker Consumptions (Mt Per Day)

- In this section, Bunker Consumption of the vessel At Sea (Ballast, Laden), At Anchorage, During Cargo Ops will be entered.

✓ BUNKER CONSUMPTIONS (MT PER DAY):

At Anchorage :	LSMGO	NA				
At Sea (Ballast) :	LSMGO	NA				
At Sea (Laden) :	LSMGO	NA				
During Cargo Ops :	LSMGO	NA				

Figure 41 Bunker Consumptions

18.3 Voyage Details

- In this section, Voyage Details of the Vessel is captured. Last Port, Arrival Port and Next Port Details will be entered by Surveyor along with LogBook Figures.

VOYAGE DETAILS

LAST PORT

Departure Date: DD-MMM-YYYY

Departure Time:

Departure Condition:

Draft Fwd:

Draft Aft:

Cargo Type/Ballast:

Cargo Quantity:

Departure ROB: LSMGO NA

ARRIVAL PORT

Arrival Date: DD-MMM-YYYY

Arrival Time:

Arrival Condition:

Draft Fwd: 0.000 m

Draft Aft: 0.000 m

ETD Date: DD-MMM-YYYY

ETD Time:

Departure Condition:

Draft Fwd: 0.000 m

Draft Aft: 0.000 m

Arrival ROB (LOG BOOK): LSMGO NA

Arrival ROB (ACTUAL): LSMGO NA

Difference: LSMGO

Departure ROB (Est): LSMGO NA

NEXT PORT

ETA DATE: DD-MMM-YYYY

ETA TIME:

Arrival Condition:

Draft Fwd:

Draft Aft:

Arrival ROB (Est): LSMGO NA

Figure 42 Voyage Details

19 Gauging Ticket:

- This is a non-editable screen. User/Surveyor can print the report and fill in the required details manually.

POOL BSOL

TTTT / 11-Jun-24 - Open

GAUGING TICKET

Name of Vessel / Bunker Tanker:

Date Of Survey:

Fwd:

Aft:

Trim:

List:

BEFORE BUNKERING

Tank No.	Grade	Ref Height (M)	OBSERVED SDG/ULL	Free Water	Temp

REMARKS

Figure 43 Gauging Ticket

20 Cover Page

- Under the Cover Page there are three sections, those are as follows: -
 - Survey Details**
 - In this section, the Survey Details are fetched from Summary Screen.
 - Summary Of Survey Details and Bunker Quantity voyage Details:**
 - In this section, the Survey Details are fetched from Summary and Bunker Quantity Details are fetched from ROB, VMR.

c) Bunker Sampling

- In this section, Sampling related information will be displayed.

d) Continuous Drip Sampling

- In this section, Drip sampling details will be displayed.

The screenshot shows the 'COVER PAGE' of the POOL BSOL system. The top navigation bar includes the text 'TTTT / 11-Jun-24 - Open' and the 'POOL BSOL' logo. The main form is divided into several sections: 'SURVEY DETAILS' with fields for Vessel (TTTT), Attending Surveyor (savita karagi), Client, Date of Attendance (11-JUN-2024), and Purpose (Bunker Quantity Survey); 'SUMMARY OF SURVEY DETAILS AND BUNKER QUANTITY' with fields for Bunker Grade (LSMGO), Supplier, Barge, Barge Officer, Initial ROB (Actual) (0.000), and Log Book ROB (0.000); 'BUNKER SAMPLING' with a dropdown for 'Was a Continuous DRIP Sample Drawn' (YES) and a text field for 'If No: What Was the Sampling Method for Bunker Samples' (NA); and 'CONTINUOUS DRIP SAMPLING' with dropdowns for 'Location of Continuous Drip Sample' (VESSEL MANIFOLD), 'Was Drip Sampler Actually Taking Samples?' (YES), and 'Was there any Attempt to Manipulate Barge Sample?' (NO). A 'Cancel' button is at the bottom left.

Figure 44 Cover Page

21 ROB ON/OFF Hire

- Rob ON/OFF Hire contains the following sub tabs.

- a) Bunker Quantity
- b) Hire Survey
- c) Vessel Hire

21.1 Bunker Quantity

- In this screen, details of bunker quantities of different Fuel Grades with presence of Chief Engineer and Logbook figures along with Flag, Port of Registry, Gross / Net Tons, Owners, Operators, Place of Survey, Survey Completed Date and Time will be displayed.
- The data for all the fields are manually entered by the Surveyors.

The screenshot shows the 'Bunker Quantity' sub-tab of the POOL BSOL system. The top navigation bar includes the text 'TTTT / 11-Jun-24 - Open' and the 'POOL BSOL' logo. The main form is divided into several sections: 'CERTIFICATE OF BUNKER QUANTITY' with fields for Flag (233), Port of Registry (455), Gross / Net Tons (56), Owners (6546), Operators (6456), and Place of Survey (6456); 'LOG BOOK FIGURES ON' with fields for Marine Fuel Oil (876.000 Mt), Very Low Sulphur Fuel Oil (34.000 Mt), Marine Gas Oil (45.000 Mt), and Low Sulphur Gas Oil (65.000 Mt); and 'REMARKS' with a 'Cancel' button. The form also includes a section for 'Survey Completed Date' (24-MAY-2024) and 'Survey Completed Time' (Invalid date). A statement at the bottom reads: 'The undersigned as independent marine surveyors, acting on behalf of (6456) carried out a Bunker Survey on (24-MAY-2024) in order to determine the quantity of bunkers remained on board.'

Figure 45 Bunker Quantity

21.2 Hire Survey

- In this screen, Time log events along with the Date and Time will be displayed.
- Surveyor can manually select Hire Certificate Type, Vessel Name, Location, Date and Grades.

Operation	Date	Time
1 SURVEYOR LEFT BASE	DD-MMM-YYYY	
2 SURVEYOR ONBOARD VESSEL	DD-MMM-YYYY	
3 COMMENCED SURVEY	DD-MMM-YYYY	
4 COMPLETED SURVEY	DD-MMM-YYYY	
5 SURVEYOR LEFT VESSEL	DD-MMM-YYYY	
6 SURVEYOR RETURN TO BASE	DD-MMM-YYYY	

Figure 46 Hire Survey

21.3 Vessel Hire:

- In this screen, details of onboard bunker quantities of Fuel Oil, Diesel/Gas Oil , Fresh Water, and Lube Oil Total values along with Vessel Name, Port, Official Number, Location, Date and Time will be displayed.
- Draft Aft, Drat Forward, Trim and Heel data will be entered by Surveyors.

Forward	Aft	Trim	Heel
11.00 m	23.00 m	34.00 m	54.00 m

Figure 47 Vessel Hire

22 Final report

- Based on the Delivery type final report contains the different tabs.
- If the delivery type is **MF** it has following tabs and they are: -

- **MFM Report**
- **Quantity Measurement Details**
- **Sampling and Office Details**

Figure 48 Final Report – MFM Tab details

22.1 MFM Report

- Under **MFM tab** there are two sections, those are as follows: -
 - Total Sealing Locations, Total Survey Time Invoice**
 - Density, Viscosity & Water Results**

22.1.1 Total sealing locations, Total Survey Time Invoice

- Under this tab, there will be Total sealing locations field and Total survey time invoice that can be entered Manually.

Figure 49 Report number, Total sealing location and Total Survey Time Invoice

22.1.2 Density, Viscosity & Water Results

- This tab contains the following fields that are:
- Lab Result Number, Density: Result, Limit, Water: Result, Limit, Kinematic Viscosity: Result, Limit (here there are two option to select one is Kinematic Viscosity 40 ° cmm²/s and one more Kinematic Viscosity 50 ° cmm²/s), and data can be entered/Selected manually as shown in the following screen.

Figure 50 Final Report – Density, Viscosity and Water results details

22.2 Quantity Measurement Details

- Under Quantity measurement details there are Three Headers, those are as follows: -
 - Bunker Ticket Number**

b) Verification of the Batch Delivered by Mass Flow Meter
c) Quantity Measurement

22.2.1 Bunker Ticket Number

- Under this tab, there will be Bunker Ticket Number field that can be entered Manually.

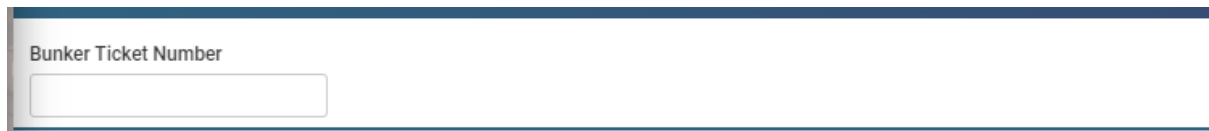
A screenshot of a software interface showing a label 'Bunker Ticket Number' above a single-line text input field. The interface has a dark blue header bar and a light gray background.

Figure 51

Figure 52 Final Report – Bunker ticket number

22.2.2 Verification Of the Batch Delivered by Mass Flow Meter

- This tab contains the following fields that are:
- Agreed Pumping Rate as per Bunker Requisition Form, Average Pumping Rate by the Mass Flow Meter, Average Receiving Rate by Vessel, Agreed Pumping Rate within Q min & Q max, Any Alarm Activated during the Delivery will be available, and data can be entered/Selected manually as shown in the following screen.

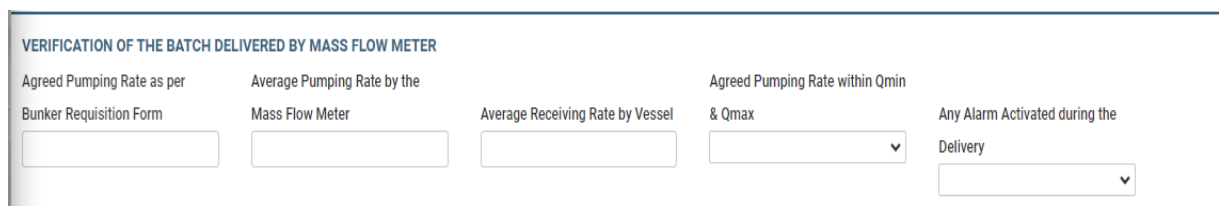
A screenshot of a software screen titled 'VERIFICATION OF THE BATCH DELIVERED BY MASS FLOW METER'. It contains five input fields arranged in two rows. The first row has: 'Agreed Pumping Rate as per Bunker Requisition Form' (text input), 'Average Pumping Rate by the Mass Flow Meter' (text input), 'Agreed Pumping Rate within Qmin & Qmax' (dropdown menu), and 'Any Alarm Activated during the Delivery' (dropdown menu). The second row has: 'Average Receiving Rate by Vessel' (text input). The interface has a light gray background and a dark blue header bar.

Figure 53 Final Report -- Verification of the Batch Delivered by Mass Flow Meter

22.2.3 Quantity Measurement

- This tab contains the following fields that are:
- For Measurement of Vessel out-turn: Volume at 15°C(m³), Density at 15°C(kg/m³), Weight factor (Density – 1.1), Weight (M tons) and data will be fetched and calculated.
- For Bunker Delivery Note: Volume at 15°C(m³),(Non editable),Density at 15°C(kg/m³) [manual entry], Weight factor (Density – 1.1), Weight (M tons) [data fetch from the summary screen survey info section BDN field].

- For Difference: Volume at 15°C(m³), Density at 15°C(kg/m³), Weight factor (Density – 1.1), Weight (M tons) [data will be calculated for all the fields].

QUANTITY MEASUREMENT				
Measurement of Vessel out-turn	Volume at 15°C(m³) 0.000	Density at 15°C(kg/m³)	Weight factor (Density – 1.1)	Weight (M tons)
Bunker Delivery Note	Volume at 15°C(m³) NA	Density at 15°C(kg/m³)	Weight factor (Density – 1.1) NA	Weight (M tons) 0.000
Difference	Volume at 15°C(m³) NA	Density at 15°C(kg/m³) 0.0	Weight factor (Density – 1.1) NA	Weight (M tons) 0.000

Figure 54 Quantity Measurement details

22.3 Sampling and Office Details

22.3.1 Sampling Information and witnessing

- This tab contains the Inspection By dropdown field and data will be selected manually.
- The table data fetch from the Sampling info screen.

SAMPLING INFORMATION AND WITNESSING					
Inspection By					
Sample	Retention Purpose	Surveyor	Bunker Tanker	Ship	Other(Specify)
1	SHIP RETENTION	NA	NA	NA	NA
2	SHIP MARPOL RETENTION	NA	NA	NA	NA
3	BUNKER TANKER RETENSION	NA	NA	NA	NA
4	BUNKER TANKER RETENSION	NA	NA	NA	NA
5	BUNKER TANKER RETENSION	NA	NA	NA	NA
6	SURVEYOR 1 RETENTION	NA	NA	NA	NA
7	SURVEYOR 2 RETENTION	NA	NA	NA	NA
8	SHIP'S LAB RETENTION	NA	NA	NA	NA

Figure 55 Sampling information and witnessing -- Inspection By details

22.3.2 Sample Remarks and Report Number

- Sampling remarks is the free text area, user can enter the data manually.
- In the Report Number field data will be selected manually.

SAMPLE REMARKS	
Report Number	

Figure 56 Sampling Remarks and report number

22.3.3 Office Information

- This tab contains the Caption Name and Designation dropdown fields and data will be selected manually.



The form is titled "OFFICE INFORMATION". It contains two dropdown menus. The first is labeled "Name" and has "Captain:" written next to it. The second is labeled "Designation". Both dropdowns have a downward arrow icon.

Figure 57 Office information -- captain details

- If the delivery type is NON MFM it has following tabs and they are: -

- **NON MFM Report**
- **Quantity Measurement**
- **Sampling and Office Details**



The screenshot shows a software interface for a "Final Report 1". At the top, it says "POOL BSOL". Below this, there are three tabs: "NON-MFM REPORT - LSMGO", "QUANTITY MEASUREMENT DETAILS", and "SAMPLING AND OFFICE DETAILS". The "NON-MFM REPORT - LSMGO" tab is selected. There are "Save" and "Cancel" buttons at the bottom.

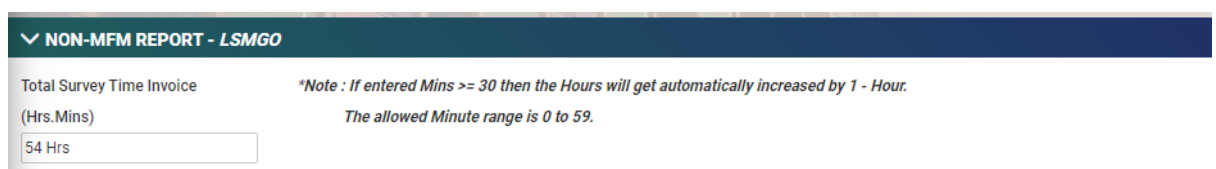
Figure 58 Final Report-Non MFM tabs details

22.4 NON MFM Report

- Under **NON MFM tab** there are two sections, those are as follows: -
 - Total Survey Time Invoice (Hrs. Mins)**
 - Opening ROB**
 - Density, Viscosity & Water Results**

22.4.1 Total Survey Time Invoice (Hrs. Mins)

- Total survey invoice time (Hrs. Mins) [manual entry, if entered Mins $\geq$ 30 then the Hours will get automatically increased by 1 - Hour]



The screenshot shows the "NON-MFM REPORT - LSMGO" section. It contains a label "Total Survey Time Invoice (Hrs.Mins)" and a text input field with "54 Hrs". To the right, there is a note: "*Note : If entered Mins $\geq$ 30 then the Hours will get automatically increased by 1 - Hour. The allowed Minute range is 0 to 59."

Figure 59 Non MFM- Total Survey invoice time

22.4.2 Opening ROB

OPENING ROB		
Requested Pumping Rate by Vessel	Average Receiving Rate by Vessel	Avg. Pumping Rate by Bunker Tanker
55 MT/Hour		

Figure 60 Non MFM- Opening ROB

22.4.3 Density, Viscosity & Water Results

- This tab contains the following fields that are:
- Lab Result Number, Density: Result, Limit, Water: Result, Limit, Kinematic Viscosity: Result, Limit (here there are two option to select one is Kinematic Viscosity 40 ° cmm²/s and one more Kinematic Viscosity 50 ° cmm²/s), and data can be entered/Selected manually as shown in the following screen.

DENSITY, VISCOSITY & WATER RESULTS					
Lab Result Number					
Density:	Result	Limit	Water:	Result	Limit
Kinematic Viscosity :					
☐	Result	Limit			
40° C mm ² /s					
☐					
50° C mm ² /s					

☐ There was a volume difference of 0.000 m³ between Vessel and Bunker Tanker. However, Chief Engineer signed the BDN quantity at MT under protest. Please refer Statement of Fact & Note of Protest for further details.

☐ The water content of % v/v translates to volume at 15 °C of 0.000 (m³). This is equivalent to 0.000 M tons of oil. This has not been deducted from the quantity calculation.

Figure 61 Non MFM – Density, Viscosity and Water results details

22.5 Quantity Measurement

- This tab contains the following fields that are:
- For Bunker Delivery Note[A]: Volume at 15°C(m³) [manual entry], Density at 15°C(kg/m³) [manual entry], Weight factor (Density – 1.1), Weight (M tons) [data will be calculated].
- For Measurement of Bunker Tanker out-turn[B]: Volume at 15°C(m³), Density at 15°C(kg/m³), Weight factor (Density – 1.1), Weight (M tons) and data will be fetched and calculated.
- For Measurement of Vessel out-turn[C]: Volume at 15°C(m³), Density at 15°C(kg/m³), Weight factor (Density – 1.1), Weight (M tons) and data will be fetched and calculated.
- For Difference[B-A]: Volume at 15°C(m³), Density at 15°C(kg/m³), Weight factor (Density – 1.1), Weight (M tons) [data will be calculated for all the fields].
- For Difference[C-B]: Volume at 15°C(m³), Density at 15°C(kg/m³), Weight factor (Density – 1.1), Weight (M tons) [data will be calculated for all the fields].
- For Difference[C-A]: Volume at 15°C(m³), Density at 15°C(kg/m³), Weight factor (Density – 1.1), Weight (M tons) [data will be calculated for all the fields].

QUANTITY MEASUREMENT DETAILS				
Bunker Delivery Note [A]	Volume at 15°C(m³)	Density at 15°C(kg/m³)	Weight factor (Density – 1.1)	Weight (M tons)
measurement of Bunker Tanker out-turn [B]	0.000			
Measurement of Vessel out-turn [C]	0.000			
Difference [B-A]	0.000	0.0	0.0	0.000
Difference [C-B]	0.000	0.0	0.0	0.000
Difference [C-A]	0.000	0.0	0.0	0.000

Figure 62 Non MFM -- Quantity Measurement details

22.5.1 Volume & Density - Gain/Loss

- This tab contains the following fields that are:
- For Volume: Unit (MT) [data will be calculated], Profit [manual Selection], Reason [manual Selection].
- For Density: Unit (kg/m³) [data will be calculated], Profit [manual Selection], Reason [manual Selection].
- Total will be calculated automatically.

VOLUME & DENSITY - GAIN/LOSS			
Volume	Unit (MT)	Profit	Reason
	0.000		
Density	Unit (kg/m³)	Profit	Reason
	0.000		
Total	0.000		

Figure 63 Volume & Density - Gain/Loss

22.6 Sampling and Office Details

22.6.1 Sampling Information and witnessing

- This tab contains the Inspection By dropdown field and data will be selected manually.
- Table data fetch from the sampling information screen.

SAMPLING INFORMATION AND WITNESSING					
Inspection By					
Sample	Retention Purpose	Surveyor	Bunker Tanker	Ship	Other(Specify)
1	SHIP RETENTION	NA	NA	NA	NA
2	SHIP MARPOL RETENTION	NA	NA	NA	NA
3	BUNKER TANKER RETENTION	NA	NA	NA	NA
4	BUNKER TANKER RETENTION	NA	NA	NA	NA
5	BUNKER TANKER RETENTION	NA	NA	NA	NA
6	SURVEYOR 1 RETENTION	NA	NA	NA	NA
7	SURVEYOR 2 RETENTION	NA	NA	NA	NA
8	SHIP'S LAB RETENTION	NA	NA	NA	NA

Figure 64 Sampling information and witnessing -- Inspection By details

22.6.2 Sample Remarks and Report Number

- The sample remarks are the free text area, user can enter the data manually.

- Report Number is the manual entry field.



Figure 65 Sampling Remarks and report number

22.6.3 Office Information

- This tab contains the Caption Name and Designation dropdown fields and data will be selected manually.



Figure 66 Office information -- captain details

23 Master Details (Web Application)

- New records for Bunker Buyer, Trader, Supplier, Fuel Grade etc., can be created on the Survey form itself. In addition, the server version has Master Data maintenance forms.
- Users can be created by admin user under **User Management** Section. As Show in the following screen are the tabs under master details screen.



Figure 67 Master Details

23.1 Bunker Tanker(Barge)

- A long flat-bottomed boat for carrying freight on canals and rivers, either under its own power or towed by another.
- It has two tabs: -

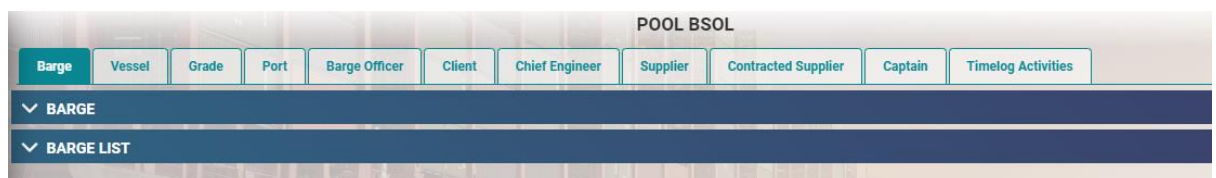


Figure 68 Master Detail-Bunker Tanker

23.1.1 Barge

- In this tab, we can add new Bunker Tanker License number, Bunker Taker Name and Status.

23.1.2 Barge List

- The added Bunker Tanker with License Number, Bunker Tanker name and Status is shown here. The table as Show in the following screen.

The screenshot shows a web interface for managing barges. At the top, there is a header bar with the title 'BARGE'. Below it, there is a form with four input fields: 'IMO Number *', 'SB No', 'Barge Name *', and 'Status'. To the right of these fields are two buttons: '+ Add' and 'Clear'. Below the form is a section titled 'BARGE LIST' with a search bar. Underneath the search bar is a table with four columns: 'IMO No', 'SB No', 'Barge Name ^', and 'Status'. The table contains five rows of data.

IMO No	SB No	Barge Name ^	Status
6453820		TEEST	Active
1564332	SB65423	TEST	Active
2312321	TEST	TEST	Active
1111111	MMTR342	TEST	Active
1235665		TTTT	Active

Figure 69 Mater Details-Bunker Tanker Details

- To add Bunker Tanker: -
- Click on Bunker Tanker under Master Details.
 - Enter Bunker Tanker License Number, Bunker Tanker Name.
 - Select the Status.
 - Click on Add.
- To Edit Bunker Tanker: -
- Select Bunker Tanker details from the list.
 - Edit the necessary changes.
 - Click on update.

23.2 Vessel

- The ship is called as Vessel.
- It has two tabs: -

The screenshot shows a web interface for managing vessels. At the top, there is a header bar with the title 'POOL BSOL'. Below it, there is a row of tabs: 'Barge', 'Vessel', 'Grade', 'Port', 'Barge Officer', 'Client', 'Chief Engineer', 'Supplier', 'Contracted Supplier', 'Captain', and 'Timelog Activities'. The 'Vessel' tab is currently selected. Below the tabs is a section titled 'VESSEL' with a sub-section 'VESSEL LIST'. Underneath 'VESSEL LIST' is a search bar.

Figure 70 Mater Details-Vessel

23.2.1 Vessel

- In this tab, we can add new vessel by entering IMO number, Vessel name and Status.

23.2.2 Vessel List

- The added vessel with IMO no., Vessel name and Status is shown here. The table as Show in the following screen

^ VESSEL

IMO Number *
Name *
Status
+ Add
Clear

^ VESSEL LIST

IMO Number	Name ^	Status
7852176	BW STREAM	Active
8523147	DO NOBIKSUKE	Active
9403102	NAVIOS VEGA	Active
8789215	NORDPENGUIN	Active
8789215	NORDPENGUIN	Active
4568795	NORDTULIP	Active
9874612	OCEANIC VIEW	Active
9654123	SAPIEN SEAS	Active
9766346	SEA HAWK	Active
9510462	SILVER POINT	Active

0 selected / 10 total

Figure 71 Master Details-Vessel Details

- To add Vessel: -
 - Click on Vessel tab under Master Details.
 - Enter IMO Number, Vessel name.
 - Select the Status.
 - Click on Add.
- To Edit Vessel: -
 - Select Vessel details from the list.
 - Edit the necessary changes.
 - Click on update.

23.3 Grade

- A particular level of rank, quality, proficiency, or value in fuel.
- It has two tabs: -

POOL BSOL

Barge
Vessel
Grade
Port
Barge Officer
Client
Chief Engineer
Supplier
Contracted Supplier
Captain
Timelog Activities

^ GRADE

^ GRADE LIST

Search

Figure 72 Master Details-Grade

23.3.1 Grade

- In this tab, we can add new grade by entering Grade name and Status.

23.3.2 Grade List

- The added vessel with Grade name and Status is shown here. The table as Show in the following screen

GRADE

Grade Name * Status + Add Clear

GRADE LIST

Grade Name ^	Status
HSFO	Active
HSGO	Active
LSMGO	Active
MDO	Active
MDO	Active
MDO 420	Active
MFO	Active
MFO 380 CST	Active
MSO	Active
SLUDGE	Active

0 selected / 10 total

Figure 73 Master Details-Grade Details

- To add Grade: -
 - Click on Grade tab under Master Details.
 - Enter the Grade name.
 - Select the Status.
 - Click on Add.
- To Edit Grade: -
 - Select Grade details from the list.
 - Edit the necessary changes.
 - Click on update.

23.4 Bunker Port

- A town or city with a harbour or access to navigable water where ships load or unload cargo.
- It has two tabs: -

POOL BSOL

Barge Vessel Grade **Port** Barge Officer Client Chief Engineer Supplier Contracted Supplier Captain Timelog Activities

▼ **PORT**

▼ **PORT LIST**

Search

Figure 74 Mater Details-Bunker Port

23.4.1 Bunker Port

In this tab, we can add new port by entering UN code, Bunker Port name and Status.

23.4.2 Bunker Port List

The added vessel with UN code, Bunker Port name and Status is shown here. The table as Show in the following screen

^ PORT

UN Code *
Port Name *
Status
+ Add
Clear

^ PORT LIST

UN Code	Port Name	Status
NLAMSTD	AMSTERDAM	Active
IN BOM	BOMBAY	Active
INBOM	BOMBAY	Active
INBOM	BOMBAY	Active
EUITL	ITALY	Active
JMJMC	JAMAICA	Active
USLSA	LOS ANGELES	Active
USNYK	NEW YORK	Active
SGSIN	SINGAPORE	Active
AUSYD	SYDNEY	Active

0 selected / 10 total

Figure 75 Mater Details-Bunker Port Details

- To add Port: -
 - Click on Port tab under Master Details.
 - Enter the UN Code and Port name.
 - Select the Status.
 - Click on Add.
- To Edit Port: -
 - Select Port details from the list.
 - Edit the necessary changes.
 - Click on update.

23.5 Barge Officer

- The one who represents barge which is provided by physical supplier is known as Cargo officer.
- It has two tabs: -

POOL BSOL

Barge Vessel Grade Port Barge Officer Client Chief Engineer Supplier Contracted Supplier Captain Timelog Activities

BARGE OFFICER

BARGE OFFICER LIST

Search

Figure 76 Master Details-Cargo Officer

23.5.1 Barge Officer

- In this tab, we can add new Cargo officer by entering Cargo officer name, and Status.

23.5.2 Barge officer List

- The added Cargo officer with Cargo Officer Name and Status is shown here. The table as Show in the following screen

^ BARGE OFFICER

Identification Number* Barge Officer* Nationality Status + Add Clear

^ BARGE OFFICER LIST

Search

Identification Number	Barge Officer ^	Nationality	Status
1111111	TESTS	INDIA	Active
UN4352UU	TESTT	INDIA	Active

Figure 77 Master Details-Cargo Officer Details

To add Barge officer: -

- Click on Cargo officer tab under Master Details.
- Enter the Cargo officer name.
- Select the Status.
- Click on Add.

To Edit Barge officer: -

- Select Cargo officer details from the list.
- Edit the necessary changes.
- Click on update.

23.6 Client:

- The company which is purchasing bunker from physical supplier is known as Customer.
- It has two tabs: -

Barge Vessel Grade Port Barge Officer **Client** Chief Engineer Supplier Contracted Supplier Captain Timelog Activities

^ CLIENT

^ CLIENT LIST

Search

Figure 78 Master Details-Customer

23.6.1 Client

- In this tab, we can add new Client by entering Client Name, Nationality, Address and Status.

23.6.2 Client List

- The added Clients details with Client name, Nationality and Status.is shown here. The table as Shown in the following screen.

CLIENT

Client Name * Nationality Status Address * + Add ↺ Clear

CLIENT LIST

Search

Client Name ^	Nationality	Status
TEST	INDIA	Active
TEST	AUSTRALIA	Active
TESTS	INDIA	Active
TESTT	INDIA	Active

Figure 79 Master Details-Customer Details

- To add Client: -
 - Click on Client tab under Master Details.
 - Enter the Client name, Nationality and Address.
 - Select the Status.
 - Click on Add.
- To edit Client: -
 - Select Client details from the list.
 - Edit the necessary changes.
 - Click on update.

23.7 Chief Engineer

- The chief engineer of the vessel who is responsible for receiving bunkers and documentation of the bunkering operation.
- It has two tabs: -

POOL BSOL

Barge Vessel Grade Port Barge Officer Client **Chief Engineer** Supplier Contracted Supplier Captain Timelog Activities

▼ CHIEF ENGINEER

▼ CHIEF ENGINEER LIST

Search

Figure 80 Master Details-Chief Engineer

23.7.1 Chief Engineer

- In this tab, we can add new Chief engineer by entering Chief engineer name, Status.

23.7.2 Chief Engineer List

- The added Chief engineer with Chief engineer name, and Status is shown here. The table as Show in the following screen

CHIEF ENGINEER

Chief Engineer Name * Status

maChiefEngg.status

CHIEF ENGINEER LIST

Chief Engineer	Status
ABCD	Inactive
ABCD	Inactive
ABCDE	Active
ABDC	Active
ADS	Inactive
ANDRIEVE SERG11	Active
AS	Active
ASKF3456	Active
AUNG NU	Active
AUNG NU	Active

0 selected / 27 total

Figure 81 Master Details-Chief Engineer Details

- To add Chief Engineer,
 - Click on Chief Engineer tab under Master Details.
 - Enter Chief Engineer Name.
 - Select Status.
 - Click on Add.
 - Chief Engineer details will be added.
- To edit the Chief Engineer details,
 - select Chief Engineer details from the list and edit.
 - After editing click on Update.

23.8 Supplier

- The company which contractually agrees with the buyer to deliver the product.
- It has two tabs: -

POOL BSOL

Barge Vessel Grade Port Barge Officer Client Chief Engineer **Supplier** Contracted Supplier Captain Timelog Activities

▼ SUPPLIER

▼ SUPPLIER LIST

Search

Figure 82 Master Details-Supplier

23.8.1 Supplier

- In this tab, we can add new Supplier by entering Supplier name, and Status.

23.8.2 Supplier List

- The added Supplier with Supplier name and Status. is shown here. The table as Shown in the following screen.

^ SUPPLIER	
Supplier Name *	Status
+ Add	Clear
^ SUPPLIER LIST	
Supplier Name ^	Status
123CHIMBUSCO PAN NATION PETRO CHEMICAL CO., LTD	Active
123CHIMBUSCO PAN NATION PETRO CHEMICAL CO., LTD	Active
ABCD	Active
ABCD	Active
ABCD1	Active
AM	Active
AM1	Active
AM1	Active
AN NATION PETRO CHEMN	Active
AN NATION PETRO CHEMN	Active
0 selected / 32 total	
Previous 1 2 3 4 Next	

Figure 83 Master Detail-Supplier Details

- To add Supplier,
 - Click on Supplier tab under Master Details.
 - Enter Supplier Name.
 - Select the status.
 - Click on Add.
 - Supplier details will be added.
- To edit the Supplier details,
 - select Supplier details from the list and edit.
 - After editing click on Update.

23.9 Contracted Supplier

- It has two tabs: -

POOL BSOL									
Barge	Vessel	Grade	Port	Barge Officer	Client	Chief Engineer	Supplier	Contracted Supplier	Captain
Contracted Supplier									
Contracted Supplier List									
Search									

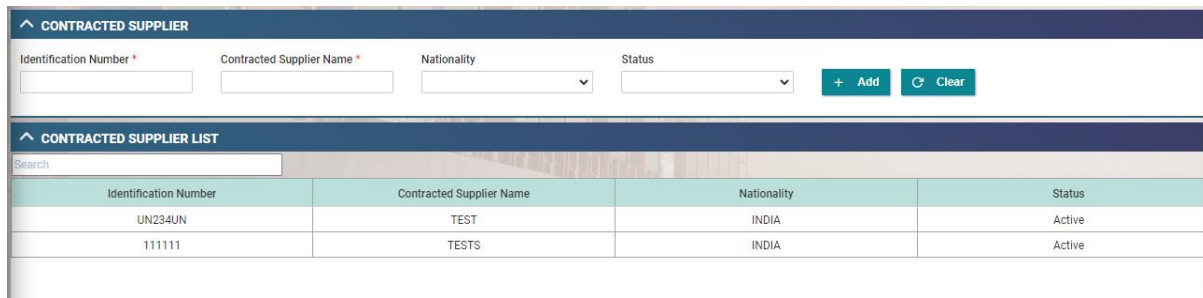
Figure 84 Master Details-Master

23.9.1 Contracted Supplier

- In this tab, we can add new Contracted Supplier by entering Identification Number, Contracted Supplier Name, Nationality and Status.

23.9.2 Contracted Supplier List

- The list of created Contracted Suppliers will be displayed here . The table as Shown in the following screen.



CONTRACTED SUPPLIER

Identification Number * Contracted Supplier Name * Nationality Status

+ Add ↺ Clear

CONTRACTED SUPPLIER LIST

Search

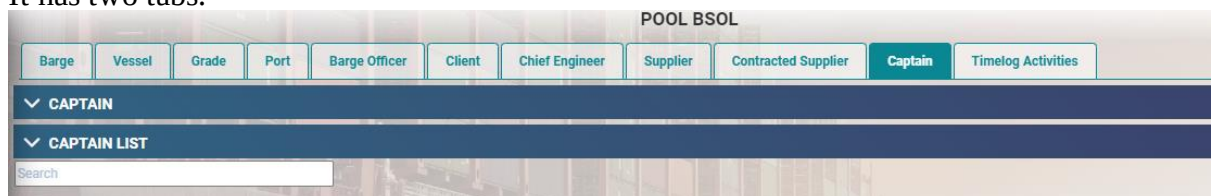
Identification Number	Contracted Supplier Name	Nationality	Status
UN234UN	TEST	INDIA	Active
111111	TESTS	INDIA	Active

Figure 85 Master Details-Master Detail

- To add Contracted Supplier,
 - Click on Contracted Supplier tab under Master Details.
 - Enter Contracted Supplier Name, Identification Number.
 - Select the Nationality.
 - Select the Status.
 - Click on Add.
 - Contracted Supplier details will be added.
- To edit the Contracted Supplier details,
 - Select Contracted Supplier details from the list and edit.
 - After editing click on Update.

23.10 Captain:

- The person in command of a ship.
- It has two tabs.



POOL BSOL

Barge Vessel Grade Port Barge Officer Client Chief Engineer Supplier Contracted Supplier **Captain** Timelog Activities

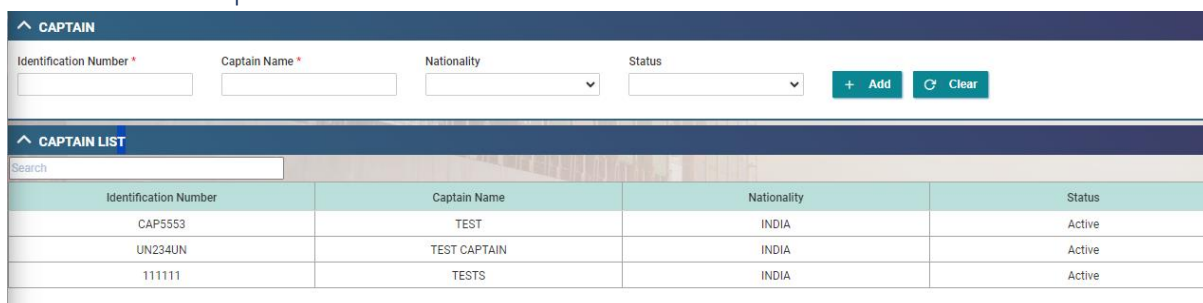
✓ CAPTAIN

✓ CAPTAIN LIST

Search

23.10.1 Captain:

23.10.2 Captain List:



^ CAPTAIN

Identification Number * Captain Name * Nationality Status

+ Add ↺ Clear

^ CAPTAIN LIST

Search

Identification Number	Captain Name	Nationality	Status
CAP5553	TEST	INDIA	Active
UN234UN	TEST CAPTAIN	INDIA	Active
111111	TESTS	INDIA	Active

23.11 Time log Activities

- The bunker surveyor shall record the time of all relevant events related to the entire bunkering operation until their return to base.

- Here, Time log Activities can be added which will be displayed in Time log screen of each survey.
- It has two tabs: -

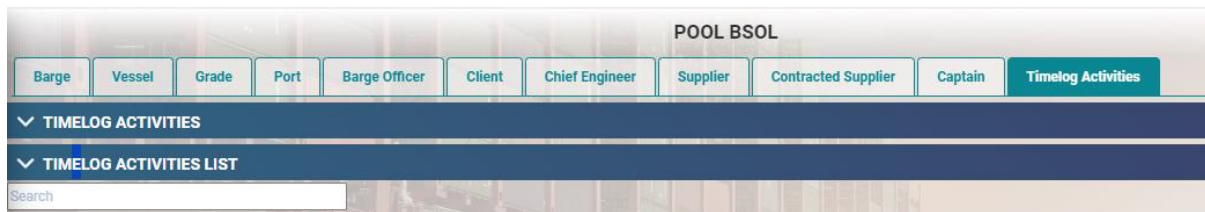


Figure 86 Master Details-Time Log Activities

23.11.1 Time log activities

- In this tab, we can add new Time log Activities by entering Activity, Display on Screen and Status.

23.11.2 Time log activities List

- The added Time log Activities with Activity, Display on Screen and Status is shown here. The table as Show in the following screen

TIMELOG ACTIVITIES		
Activity *	Display On Screen *	Status
+ Add Clear		
TIMELOG ACTIVITIES LIST		
Activity ^	Display On Screen	Status
BARGE ARRIVED	Yes	Active
BARGE DEPARTED	Yes	Active
BARGE HALT	Yes	Inactive
BUNKER TANKER ALONGSIDE	Yes	Active
BUNKER TANKER CAST-OFF	Yes	Active
CAPTAIN ARRIVED	Yes	Active
CHIEF ENGINEER ARRIVED	Yes	Active
COMMENCE PUMPING	Yes	Active
COMPLETED PUMPING	Yes	Active
DEPARTURE PILOT BOARDED	Yes	Active
0 selected / 40 total		

Figure 87 Master Details-Time Log Activities and Time Log Activity List

- To add Time log Activities:
 - Click on Time log Activities tab under Master Details.
 - Enter the Activity Name.
 - Select **Yes** from **Display on Screen** dropdown. (If selected as **No**, the activity will not be displayed in Time log screen).
 - Select the Status as **Active**. (If selected as **Inactive**, the activity will not be displayed in Time log screen).
 - Click on Add.
 - Time log Activities will be added.

- To edit the Time log Activities,
 - select Time log Activities from the list and edit.
 - After editing click on Update.

Note: -

- Once data is filled in master details for all the tabs, we can get the list of options while creating survey by just typing the 1st character of that field.
- Similarly, all other tabs can be added and editing by following the above steps.

24 Appendix

What is bunker fuel?

A combustible fuel used in the operation of ship engines, generators and boilers, the latter usually for heating the cargo material being transported. In the early days of steamships, steam was produced in boilers run on coal. The compartment where the coal was stored was known as the bunker and this term has carried through to today's fuel.



COAL BUNKER ON AN EARLY STEAMSHIP



STOKING THE SHIP BOILERS WITH COAL

Types of bunker fuels: -

- **MFO** (Marine fuel oil) – All types of bunker fuels
- **MGO** (Marine gas oil) - Roughly equivalent to No. 2 fuel oil, made from distillate only.
- **MDO** (Marine diesel oil) – Low viscosity blend of heavy gasoil that may contain very small amounts of black oil feed stocks, does not need to be heated for use in internal combustion engines.
- **IFO** (Intermediate fuel oil) A blend of gasoil and heavy fuel oil, with less gasoil than marine diesel oil.
- **HFO** (Heavy fuel oil) - Pure or nearly pure residual oil, roughly equivalent to No. 6 fuel.

Some Important Definitions: -

- **Calibration** - The process of adjusting, or of measuring the performance of a device.
- **Calibration Tables** (Gauge or Tank Tables, Innage/Ullage Tables) Tables developed by recognized industry methods that represent volumes in each tank according to the liquid (innage) or empty space (ullage) measured in the tank. The tables are entered with linear measurements (for example, feet, inches, meters, centimeters) to obtain calibrated volumes (for example, barrels, cubic meters, or cubic feet).
- **Density:** - The density of a homogeneous substance is the ratio of its mass to its volume. The density varies as the temperature changes, and it is usually expressed as the mass per unit volume at a specified temperature.
- **Free Water (FW)** - The water present in a container that is not in suspension in the contained liquid (oil).
- **Gross Observed Volume (GOV)** - The total volume of all petroleum liquids and sediment and water, excluding free water, at observed temperature and pressure.
- **Gross Standard Volume (GSV)** - The total volume of all petroleum liquids and sediment and water, excluding free water, corrected by the appropriate volume correction factor (Ctl) for the observed temperature and API (American Petroleum Institute) gravity, relative density, or density to a standard temperature such as 60°F or 15°C and also corrected by the applicable pressure correction factor (Cpl) and meter factor.
- **List** - The leaning or inclination of a vessel expressed in degrees port or starboard.
- **List Correction** - The correction applied to the observed gauge or observed volume when a vessel is listing, provided that liquid is in contact with all bulkheads in the tank. Correction for list may be made by reference to the vessels list correction tables for each tank or by mathematical calculations.
- **Reference Height** - The distance from the tank bottom and/or datum plate to the established reference point or mark.
- **Sampling** - The process of obtaining a sample of the material in the tank, container or pipeline to use for testing or other purposes.
- **Sounding-** is the measured height of the fluid from the surface of the fluid to the bottom of the tank.
- **Sludge** - A highly viscous mixture of oil, water, sediment, and residue.
- **Total Observed Volume (TOV)** -The total measured volume of all petroleum liquids, sediment and water, and free water at observed temperature and pressure.
- **Trim** - The condition of a vessel with reference to its longitudinal position in the water. It is the difference between forward and aft drafts and is expressed as vessel being by the head or by the stern.

- **Trim Correction** - The correction applied to the observed gauge or observed volume in a vessel's tank when a vessel is not on an even keel provided that the liquid is in contact with all bulkheads in the tank. Correction for trim may be made by referencing trim tables for each tank or by mathematical calculation.
- **Ullage**- is the void space in the tank measured from the top of the tank to the upper surface of the fluid.